

Boundary conditions for thermal lattice Boltzmann equation method



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ABSTRACT

We propose a thermal boundary condition treatment based on the “bounce-back” idea and interpolation of the distribution functions for both the Dirichlet and Neumann (normal derivative) conditions in the thermal lattice Boltzmann equation (TLBE) method. The coefficients for the distribution functions involved are determined to satisfy the Dirichlet or Neumann condition with second-order accuracy. For the Dirichlet condition there is an adjustable parameter in the treatment and three particular schemes are selected for demonstration, while for the Neumann condition the second-order accurate scheme is unique. When applied to inclined or curved boundaries, the Dirichlet condition treatment can be directly used, while the Neumann condition given in the normal direction of the boundary should be converted into derivative conditions in the discrete velocity directions of the TLBE model. A spatially coupled formula relating the boundary temperature, boundary normal heat flux, and the distribution functions near the boundary is thus derived for the Neumann problems on curved boundaries. The applicability and accuracy of the present boundary treatments are examined with several numerical tests for which analytical solutions are available, including the 2-dimensional (2-D) steady-state channel flow, the 1-D transient heat conduction in an inclined semi-infinite solid, the 2-D steady-state and transient heat conduction inside a circle and the 3-D steady-state circular pipe flow. While the Dirichlet condition treatment leads to second-order accuracy for the temperature field, it only gives a first-order accurate boundary heat flux because of the irregularity of the cuts by the curved boundary with the lattices. With the Neumann condition on the curved boundary, the accuracy for the temperature field obtained is first-order. When the tangential temperature gradient on the boundary is decoupled, second-order convergence of the temperature field can be obtained with Neumann conditions.

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1. Introduction

The lattice Boltzmann equation (LBE) method has become an attractive alternative numerical method for fluid dynamics simulations because it is explicit in formulation, easy to implement and compatible with parallelization compared with the solvers for the Navier–Stokes equations. It has been successfully applied in a variety of fluid flow problems [1–4] and has been extended to complex flows, such as multi-phase and multi-component flows [5–9], turbulent flows [9–11], micro-flows [9,12], and flows through porous media [13,14].

The LBE method simulates the particle interactions on a discrete phase space with a specific set of discrete velocities, and the particle dynamics is represented by the particle distribution functions (also called particle density distribution function

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or particle velocity distribution function). The velocity distribution functions of the particles are governed by the LBE, from which the macroscopic Navier–Stokes equations can be recovered. Particle collision in the LBE method is approximated by a relaxation process toward the equilibrium state of the particle distributions. There are basically two collision models for the LBE methods: the first and most commonly used single-relaxation-time (SRT), or the Bhatnagar–Gross–Krook (BGK) model [1,15] based on the work in [16]; and the multiple-relaxation-time (MRT), also known as the moment model [17–19]. The MRT–LBE model overcomes some obvious defects of the BGK model such as fixed Prandtl number and fixed ratio between the kinematic and bulk viscosities [18] and shows better numerical stability [19] and reduced numerical dispersion [4].

Because of the advantages of the LBE method, there have been continuous interest and effort in developing thermal LBE (TLBE) models for thermal transport problems [20–28]. In general, the existing TLBE models fall into four categories: the passive scalar approach, the multispeed approach, the hybrid approach, and the double-population distribution function (DDF) approach.

In the passive-scalar-based TLBE model, temperature is considered as a passive scalar that is advected by the velocity field and satisfies the convection–diffusion equation [21]. The main shortcoming of this model is that the temperature field is driven by the flow velocity, while the temperature does not affect the flow field. The multispeed approach is a direct extension of the athermal LBE model so that only the velocity distribution function is used. In order to obtain the macroscopic temperature, additional discrete speeds and higher-order velocity expansion for the Maxwell–Boltzmann equilibrium distribution are required [22]. In the hybrid approach, the mass and momentum conservation equations are solved with the athermal LBE model, whereas the advection–diffusion equation for the temperature is solved separately by using the finite-difference method or other conventional numerical techniques [23]. The DDF approach utilizes two different distribution functions, one for the velocity field and the other one for the temperature or energy distributions. With this approach, the interaction between the flow fields (density and velocity) and the temperature field can be properly accounted for. Based on the methods for defining the temperature distribution functions, the DDF approach can be further classified into two sub-groups. The first derives the temperature distribution function from the continuous Boltzmann equation as that in athermal LBE models. Thus these two sets of distribution functions are explicitly coupled in the thermal LBE. For example, He et al. [24] used an internal-energy-based distribution function and Guo et al. [25] used a total-energy-based distribution function to simulate thermal flows. In the other sub-group of the DDF approach, however, the temperature distribution function is defined with the objective of recovering the macroscopic convection–diffusion equation (CDE), in which other terms in the energy equation such as pressure work and heat dissipation are conveniently treated as source terms. With this approach the macroscopic velocities, rather than the velocity distribution functions, are included in the evolution equation of the temperature distribution functions. Thus no explicit coupling between the velocity- and temperature-distribution functions is needed. To this end, Shi and Guo [26] proposed a thermal LBE model for nonlinear CDE simulations, in which the equilibria of the temperature distribution functions are defined to recover the correct CDE. Recognizing the advantages of the MRT model over the BGK model, Mezrhab et al. [27] and Yoshida and Nagaoka [28] proposed their MRT–TLBE models for the CDE. To recover the CDE, the equilibria of the thermal moments are specified in Mezrhab’s model [27] while the thermal equilibrium distribution functions are specified in Yoshida and Nagaoka’s model [28]. A detailed derivation of the macroscopic CDE from the MRT–TLBE model can be found in [28]. The present work also uses the MRT–LBE model by Yoshida and Nagaoka [28].

Accurate boundary condition treatment is essential to the overall integrity of the LBE simulations. For the fluid velocity field, the “bounce-back” scheme has been intensively studied [29–35] and extended to curved-boundary situations [36–40]. With appropriate modifications, some of the athermal boundary condition treatments have been extended to TLBE simulations as well. He et al. [24] extended the bounce-back rule of the non-equilibrium distributions proposed by Zou and He [34] for thermal boundary conditions. This non-equilibrium extrapolation approach was also applied by Tang et al. [41], Huang et al. [42] and Jeong et al. [43]. Liu et al. [44], however, used correction functions for the unknown energy distributions based on local known energy distributions, which is an extension of their hydrodynamic boundary condition treatment [45]. It is noted that the thermal boundary condition treatments in [41–44] were directly proposed for Dirichlet boundary conditions and the Neumann (flux) condition was transformed into Dirichlet condition by various finite-difference schemes. This transformation will become inconvenient when mixed thermal boundary conditions are encountered. As a result, there naturally is a need for a direct boundary condition treatment for the Neumann condition. D’Orazio et al. [46] introduced a “counter-slip” thermal energy density to represent the given temperature or heat flux condition on the boundary. Kuo and Chen [47] proposed a non-equilibrium “mirror-reflection” scheme to implement both the isothermal and adiabatic thermal boundary conditions. Ginzburg [48] extended the “multi-reflection” approach proposed in [32] to model both Dirichlet and Neumann boundary conditions. In the 2-D MRT–TLBE model by Mezrhab et al. [27], the bounce-back scheme was implemented to treat Dirichlet boundary conditions, where they provided an explicit formula to determine the unknown populations based on the distributions at the neighboring fluid nodes and the temperature on the boundary; and a simple “mirroring” treatment was used for the adiabatic condition. In the MRT–TLBE model by Yoshida and Nagaoka [28] for both 2-D and 3-D cases, boundary condition treatments for both the Dirichlet and the Neumann conditions were provided for the “halfway” case where the physical boundary is placed in the center of the lattice cell.

Although thermal boundary condition treatment is an important issue, the discussions about the applicability and accuracy of these treatments in curved-boundary situations in the literature are rather limited. Huang et al. [42] extended the non-equilibrium approach to treat curved boundary conditions, where the equilibrium part of the unknown population is evaluated according to the Dirichlet condition and linear extrapolation is used for the non-equilibrium component; for

Neumann conditions, they also transferred them into Dirichlet conditions with a finite difference scheme and a simple projection rule was used to convert the normal boundary flux into the heat fluxes in the discrete velocity directions of the TLBE model. Second-order accuracy of the temperature field was observed for a Dirichlet condition on the curved boundary while no examples were provided regarding their heat flux conversion and the accuracy of the transferred Neumann condition treatment in their work. For curved boundaries with Neumann conditions, Yoshida and Nagaoka [28] realized that the “half-way” thermal bounce-back scheme leads to considerable disagreement between their simulation and analytical solutions. Thus they presented a modified scheme with the intersection area accounted for to alleviate the discrepancy. Noticeable improvement was achieved in their numerical example but the accuracy was not investigated in detail. In the “multi-reflection” approach for arbitrarily shaped surfaces, Ginzburg [48] decomposed the equilibrium populations into the symmetric and anti-symmetric components. The symmetric part was tuned to build second-order and “third-order” accurate Dirichlet boundary conditions and the anti-symmetric part was used for Neumann conditions. It was also pointed out in [48] that the Neumann condition with zero tangential heat flux yielded second-order accuracy while only linear convergence was achieved when non-zero tangential heat flux exists. It is also noted that the heat flux in one of the discrete velocity directions is required in [48] in order to remove the tangential flux constraint and to obtain linear convergence. In principle, however, only the derivative in the normal direction is specified in Neumann problems. Reference [48] did not provide a means to evaluate the heat fluxes in the discrete lattice velocity directions based on the normal derivative on the curved boundary.

The objectives of the present work are twofold. The first one is to develop second-order accurate boundary condition treatments for both the Dirichlet and Neumann conditions on straight boundaries that can be placed at arbitrary locations in the lattice. The second is to extend the present boundary condition treatments into curved-boundary situations and to investigate the effect of curved-geometry on the accuracy of the present boundary condition treatments.

The present boundary condition treatments are independent of the specific TLBE model and the collision operator; thus for illustration purposes, we formulate the problem using Yoshida and Nagaoka’s MRT-TLBE model [28]. For the Dirichlet condition, there is an adjustable parameter in the treatment with second-order accuracy. Three particular choices are discussed and their performances are examined. For the Neumann problem, the second-order accurate boundary scheme is unique. When the present boundary treatments are extended to inclined or curved boundary walls, the Dirichlet condition can be directly implemented. For Neumann problems on curved boundaries the given heat flux condition in the normal direction alone cannot be used to determine the fluxes in the discrete velocity directions in the TLBE model. Only for the special case where the tangential temperature gradient does not contribute to the fluxes in the discrete directions, i.e., the tangential gradient is known to be zero or the boundary is perpendicular to one of the discrete velocity directions, can a simple formula for the distribution function be obtained with second order accuracy based on the normal derivative. For general cases the boundary heat fluxes in the discrete velocity directions must be coupled with the unknown temperature value on the boundary which can be independently evaluated on each lattice velocity direction. An expression for the heat flux determination in the discrete velocity directions is derived from the solution of the coupled equations. The accuracy of the proposed boundary condition treatments is examined using a series of numerical tests for which analytical solutions are available. The test cases include: (i) 2-D steady-state channel flow with a constant and uniform velocity and a sinusoidal wall temperature or heat flux boundary condition; (ii) 1-D transient heat conduction in an inclined semi-infinite solid with a constant normal heat flux at the end; (iii) 2-D steady-state heat conduction inside a circle with a sinusoidal wall temperature or heat flux boundary condition; (iv) 2-D transient heat conduction inside a circle with a transient wall temperature boundary condition; and (v) 3-D steady-state circular pipe flow with a constant and uniform velocity and a sinusoidal wall temperature or heat flux boundary condition. Because of the irregularity of the lattice fractions cut by the curved boundary, second-order accuracy for the temperature field is realized for Dirichlet problems, but only first-order accuracy can be obtained for Neumann problems on curved boundaries.

2. Thermal LBE method

2.1. The convection–diffusion equation

Following [28], the standard convection–diffusion equation (CDE) can be written as

$$\frac{\partial \phi}{\partial t} + \frac{\partial}{\partial x_j} (v_j \phi) = \frac{\partial}{\partial x_i} \left(D_{ij} \frac{\partial \phi}{\partial x_j} \right) + G, \quad (1)$$

where ϕ is a scalar variable that can be either temperature in thermal simulations or mass concentration in mass transfer problems, v_j is the velocity component in the x_j direction, D_{ij} is the diffusion coefficient, and G represents the source term.

The Dirichlet boundary condition is

$$\phi = \Phi_d, \quad (2)$$

and for the Neumann boundary condition involving normal derivative, it is in general written as

$$-n_i D_{ij} \frac{\partial \phi}{\partial x_j} + n_j v_j \phi = \Phi_n, \quad \text{Robin 边界条件一般格式} \quad (3)$$

where Φ_d and Φ_n are given functions of time and space on the boundary, n_i and n_j denote the i -th and j -th components, respectively, of the unit normal vector pointing inward to the domain.

2.2. Thermal LBE model

The discrete phase space in the TLBE model is defined by a regular lattice in D dimensions together with a set of discrete velocities $\{\mathbf{e}_\alpha | \alpha = 0, 1, \dots, m\}$. Analogous to the athermal MRT–LBE model, Yoshida and Nagaoka [28] proposed a D3Q7 thermal LBE model (Fig. 1) with an MRT collision operator for the CDE, in which they have carried out an asymptotic analysis based on diffusive scaling to show that the TLBE model is first- and second-order accurate in time and space, respectively. This MRT–TLBE model is also used in the present work.

The TLBE governs the set of temperature distribution functions $\{g_\alpha(\mathbf{x}, t) | \alpha = 0, 1, \dots, m\}$, from which the leading-order solution of the macroscopic temperature is obtained with $\phi = \sum_{\alpha=0}^m g_\alpha$. In the D3Q7 TLBE model, the discrete velocity set is defined as

$$\mathbf{e}_\alpha = \begin{cases} (0, 0, 0) & (\alpha = 0), \\ (\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1), & (\alpha = 1, 2, 3, 4, 5, 6), \end{cases} \quad (4)$$

The evolution of the TLBE can be written as

$$g_\alpha(\mathbf{x} + \mathbf{e}_\alpha \delta t, t + \delta t) - g_\alpha(\mathbf{x}, t) = \mathbf{L} g_\alpha(\mathbf{x}, t) + \omega_\alpha G \delta t, \quad (5)$$

where δt is the time step, ω_α is the weight coefficient defined to satisfy the CDE in Eq. (1), and $\mathbf{L}$ is the collision operator.

In the SRT–BGK model, the collision operator is approximated by

$$\mathbf{L} g_\alpha(\mathbf{x}, t) = -\frac{1}{\tau} (g_\alpha(\mathbf{x}, t) - g_\alpha^{\text{eq}}(\mathbf{x}, t)), \quad (6)$$

where τ is a coefficient that represents the relaxation time relative to the time step. In the BGK thermal model by Yoshida and Nagaoka [28], the following equilibrium distribution functions are proposed to recover the CDE:

$$g_\alpha^{\text{eq}}(\mathbf{x}, t) = \left(\omega_\alpha + \frac{\delta t v_j}{\delta x \varepsilon_D} e_{\alpha j} \omega_\alpha \right) \phi(\mathbf{x}, t), \quad (7)$$

where δx is the grid spacing in the lattice. The weight coefficients are set to be

$$\omega_\alpha = \begin{cases} 1/4, & (\alpha = 0), \\ 1/8, & (\alpha = 1, 2, 3, 4, 5, 6), \end{cases} \quad (8)$$

and the constant $\varepsilon_D = 1/4$ for the 3-D model.

In the MRT model [28], the collision operator is given by

$$\mathbf{L} g_\alpha(\mathbf{x}, t) = \mathbf{M}^{-1} \mathbf{S} \mathbf{M} \mathbf{Q} g_\alpha(\mathbf{x}, t). \quad (9)$$

Thus the evolution of the TLBE becomes

$$g_\alpha(\mathbf{x} + \mathbf{e}_\alpha \delta t, t + \delta t) - g_\alpha(\mathbf{x}, t) = \mathbf{M}^{-1} \mathbf{S} \mathbf{M} \mathbf{Q} g_\alpha(\mathbf{x}, t) + \omega_\alpha G \delta t, \quad (10)$$

and the matrix $\mathbf{Q}$ is explicitly given as below based on Eq. (7)

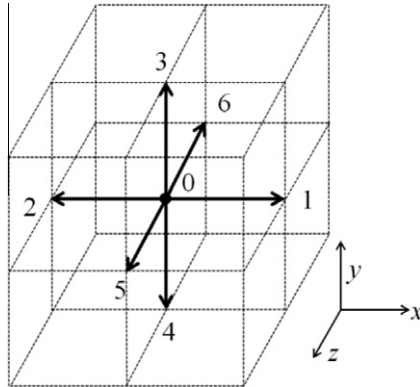


Fig. 1. Discrete velocity set $\{\mathbf{e}_\alpha\}$ for the D3Q7 TLBE model.

$$\mathbf{Q} = \left[(\omega_0, \omega_1, \dots, \omega_6)^T + \frac{\delta t v_j}{\delta x \varepsilon_D} (e_{0j}\omega_0, e_{1j}\omega_1, \dots, e_{6j}\omega_6)^T \right] (1, 1, 1, 1, 1, 1, 1) - \mathbf{I}_{7 \times 7}, \quad (11)$$

with $\mathbf{I}_{7 \times 7}$ being the 7×7 identity matrix. The transformation matrix $\mathbf{M}$ for the D3Q7 TLBE model was given in [28] as

$$\mathbf{M} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 6 & -1 & -1 & -1 & -1 & -1 & -1 \\ 0 & 2 & 2 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 1 & -1 & -1 \end{bmatrix}. \quad (12)$$

and the relaxation matrix $\mathbf{S}$ is

$$\mathbf{S}^{-1} = \begin{bmatrix} \tau_0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \tau_{xx} & \tau_{xy} & \tau_{xz} & 0 & 0 & 0 \\ 0 & \tau_{xy} & \tau_{yy} & \tau_{yz} & 0 & 0 & 0 \\ 0 & \tau_{xz} & \tau_{yz} & \tau_{zz} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \tau_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \tau_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \tau_6 \end{bmatrix}. \quad (13)$$

As pointed out in [28], the off-diagonal components in the relaxation matrix can take non-zero values to account for anisotropic diffusion. The relaxation coefficients are related to the diffusion coefficient matrix as

$$\tau_{ij} = \frac{1}{2} \delta_{ij} + \frac{\delta t}{\varepsilon_D (\delta x)^2} D_{ij}, \quad (14)$$

in order to recover the solution of the CDE to the leading-order. In the above, δ_{ij} is the Kronecker's delta.

As in the athermal MRT-LBE model, the equilibria of the conserved “hydrodynamic” moments and the non-conserved “kinetic” moments are explicitly defined [18]; the equilibria of the conserved “thermal” moments and the non-conserved “kinetic” moments $\{m_\alpha^{\text{eq}} | \alpha = 0, 1, \dots, m\}$ in the MRT-TLBE model can also be explicitly specified, as demonstrated in the D2Q5 model by Mezrhab et al. [27]. In the following we show that the D3Q7 TLBE model in [28] can be equivalently formulated with the equilibrium moments explicitly defined.

The thermal LBE is rewritten as

$$g_\alpha(\mathbf{x} + \mathbf{e}_\alpha \delta t, t + \delta t) - g_\alpha(\mathbf{x}, t) = -\mathbf{M}^{-1} \mathbf{S} (m_\alpha(\mathbf{x}, t) - m_\alpha^{\text{eq}}(\mathbf{x}, t)) + \omega_\alpha G \delta t, \quad (15)$$

where the transformation matrix $\mathbf{M}$ and the relaxation matrix $\mathbf{S}$ are the same as that in Eqs. (12) and (13), respectively; the thermal moments $m_\alpha(\mathbf{x}, t)$ are related to the distribution functions $g_\alpha(\mathbf{x}, t)$ by

$$\mathbf{m} = \mathbf{M} \mathbf{g}, \quad (16)$$

and the equilibria of the moments are defined as

$$\mathbf{m}^{\text{eq}} = (m_0^{\text{eq}}, m_1^{\text{eq}}, m_2^{\text{eq}}, m_3^{\text{eq}}, m_4^{\text{eq}}, m_5^{\text{eq}}, m_6^{\text{eq}})^T = (0, uT, vT, wT, aT, 0, 0)^T, \quad (17)$$

where a is a constant related to the coefficients ω_α by

$$a = (7\omega_0 - 1) = 3/4. \quad (18)$$

Now the conservation of the first moment results in the macroscopic temperature

$$m_0 = T = \sum_{\alpha=0}^6 g_\alpha. \quad (19)$$

Comparing Eq. (15) with Eq. (10), we can relate these two equivalent forms of evolution by

$$m_\alpha(\mathbf{x}, t) - m_\alpha^{\text{eq}}(\mathbf{x}, t) = -\mathbf{M} \mathbf{Q} g_\alpha(\mathbf{x}, t). \quad (20)$$

Similarly, keeping only the first five discrete velocity components from Fig. 1 for the D2Q5 TLBE model yields the following from Eq. (8)

$$\mathbf{e}_\alpha = \begin{cases} (0, 0), & (\alpha = 0), \\ (\pm 1, 0), (0, \pm 1), & (\alpha = 1, 2, 3, 4), \end{cases} \quad (21)$$

and the transformation from the distribution functions g_α to the thermal moments m_α now becomes

$$\mathbf{m} = \begin{bmatrix} m_0 \\ m_1 \\ m_2 \\ m_3 \\ m_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ 4 & -1 & -1 & -1 & -1 \\ 0 & 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ g_3 \\ g_4 \end{bmatrix} = \mathbf{M}\mathbf{g}. \quad (22)$$

The corresponding equilibria of the moments are defined as

$$\mathbf{m}^{\text{eq}} = (m_0^{\text{eq}}, m_1^{\text{eq}}, m_2^{\text{eq}}, m_3^{\text{eq}}, m_4^{\text{eq}})^T = (0, uT, vT, aT, 0)^T. \quad (23)$$

In the D2Q5 model, the weight coefficients are chosen to be

$$\omega_\alpha = \begin{cases} 1/3, & (\alpha = 0), \\ 1/6, & (\alpha = 1, 2, 3, 4), \end{cases} \quad (24)$$

so that the constant ε_D in Eq. (7) is $\varepsilon_D = 1/3$, and the constant a in Eq. (23) is determined by

$$a = (5\omega_0 - 1) = 2/3. \quad (25)$$

The relaxation matrix $\mathbf{S}$ for the D2Q5 model now becomes

$$\mathbf{S}^{-1} = \begin{bmatrix} \tau_0 & 0 & 0 & 0 & 0 \\ 0 & \tau_{xx} & \tau_{xy} & 0 & 0 \\ 0 & \tau_{xy} & \tau_{yy} & 0 & 0 \\ 0 & 0 & 0 & \tau_3 & 0 \\ 0 & 0 & 0 & 0 & \tau_4 \end{bmatrix}, \quad (26)$$

and the conservation of the first moment results in the macroscopic temperature

$$m_0 = T = \sum_{\alpha=0}^4 g_\alpha. \quad (27)$$

For efficient computation, the TLBE is solved in two steps:
collision step

$$\hat{g}_\alpha(\mathbf{x}, t) = g_\alpha(\mathbf{x}, t) - \mathbf{M}^{-1} \mathbf{S} (m_\alpha(\mathbf{x}, t) - m_\alpha^{\text{eq}}(\mathbf{x}, t)) + \omega_\alpha G \delta t, \quad (28)$$

and

streaming step

$$g_\alpha(\mathbf{x} + \mathbf{e}_\alpha \delta t, t + \delta t) = \hat{g}_\alpha(\mathbf{x}, t), \quad (29)$$

where $\hat{g}_\alpha$ represents the post-collision state.

2.3. Boundary conditions

The lattice cut by the boundary wall with arbitrary intersection distance is illustrated in Fig. 2. In the present work, the lattice spacing is set to be unity ($\delta x = \delta y = 1$) and the fraction of the intersected link in the field to be solved is denoted by $\Delta = ||\mathbf{x}_f - \mathbf{x}_w|| / ||\mathbf{x}_f - \mathbf{x}_e||$ with $0 \leq \Delta < 1$. Boundary condition treatment for the TLBE simulation is required during the streaming step when unknown populations from the undefined exterior nodes ($\mathbf{x}_e$ in Fig. 2) to the “field” nodes ($\mathbf{x}_f$) are encountered. It should be noted that the CDE can also be used to govern the energy equation in the solid phase by setting the velocity to zero; thus the TLBE method can be extended to simulate the thermal field in the solid phase. Hence we replace the conventional “fluid” node with the more general “field” node notation in the computational domain.

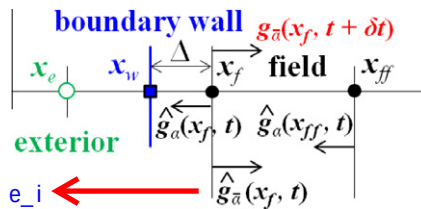


Fig. 2. Schematic depiction of the lattice link intersected by a boundary wall.

The standard momentum “bounce-back” scheme is derived for the case where the distance from the boundary wall to the nearest field lattice node in the computational domain is equal to half of the lattice spacing. The basic idea of the momentum bounce-back can be applied to derive the thermal boundary conditions in TLBE. Considering the special case of $\Delta = 0.5$, Yoshida and Nagaoka [28] proposed the following treatments for the Dirichlet and Neumann boundary conditions for walls perpendicular to one of the lattice velocity directions in the D3Q7 model. An asymptotic analysis was also given in their work to show that second-order accuracy is obtained for both conditions.

Dirichlet condition:

$$g_{\alpha}(\mathbf{x}_f, t + \delta t) = -\hat{g}_{\alpha}(\mathbf{x}_f, t) + \varepsilon_D \Phi_d, \quad (30)$$

Neumann condition:

$$g_{\alpha}(\mathbf{x}_f, t + \delta t) = \hat{g}_{\alpha}(\mathbf{x}_f, t) + (\delta t / \delta x) \Phi_n, \quad (31)$$

where $\mathbf{n}$ is parallel to $\mathbf{e}_{\alpha}$ and $\mathbf{e}_{\alpha}$. If $\mathbf{n}$ is not parallel to $\mathbf{e}_{\alpha}$, one would need to replace Φ_n in Eq. (31) with $\Phi_{n\bar{\alpha}}$, the heat flux in $\mathbf{e}_{\bar{\alpha}}$ direction which in turn involves the unknown heat flux in the tangential direction normal to $\mathbf{e}_{\alpha}$.

It is important to have second-order accurate boundary conditions for the Dirichlet and Neumann problems for arbitrary Δ values. Based on the bounce-back idea on the boundary and spatial interpolation, one can develop various boundary schemes with different number of neighboring field nodes and their populations in different discrete directions. The schemes in Eqs. (30) and (31) above use only one post-collision value $\hat{g}_{\alpha}(\mathbf{x}_f, t)$ at the node adjacent to the boundary in $\mathbf{e}_{\alpha}$ direction; thus it is considered as a “localized scheme”. The “multi-reflection” schemes in [48] utilize five post-collision neighboring values involving three field nodes, and thus up to “third-order” accurate boundary condition schemes were derived for the Dirichlet condition. In this work we show that second-order accuracy for both the Dirichlet and Neumann boundary conditions can be achieved by using only three post-collision neighboring values, $\hat{g}_{\alpha}(\mathbf{x}_f, t)$, $\hat{g}_{\alpha}(\mathbf{x}_{ff}, t)$, and $\hat{g}_{\alpha}(\mathbf{x}_f, t)$ for each unknown population $g_{\alpha}(\mathbf{x}_f, t + \delta t)$. Furthermore, we compare the present second-order Dirichlet boundary condition treatment with the “third-order” scheme in [48] by examining the errors of the temperature fields, the boundary and interior heat fluxes with numerical examples and observe that: (i) both show quadratic convergence since the TLBE model is second-order accurate in the interior space; (ii) the “third-order” scheme yields close, quantitatively even higher for some cases, numerical errors comparing with the present treatment under the same conditions; (iii) both give very similar performance for curved-boundary simulations such as quadratic convergence of the temperature field and linear convergence of the heat flux on the boundary; thus the degradation of the accuracy due to the curved-boundary cannot be mitigated by increasing the order-of-accuracy on the boundary by using more lattice nodes and populations in the field. The construction of the present boundary condition treatment is given next in detail.

2.3.1. Dirichlet boundary condition

As shown in Fig. 2, the present treatment for the Dirichlet boundary condition is given by

$$g_{\alpha}(\mathbf{x}_f, t + \delta t) = c_{d1} \hat{g}_{\alpha}(\mathbf{x}_f, t) + c_{d2} \hat{g}_{\alpha}(\mathbf{x}_{ff}, t) + c_{d3} \hat{g}_{\alpha}(\mathbf{x}_f, t) + c_{d4} \varepsilon_D \Phi_d, \quad (32)$$

where c_{d1} , c_{d2} , c_{d3} , and c_{d4} are coefficients related to the specific Δ value. To maintain second-order accuracy, the following relations are required, leaving c_{d1} adjustable (see Eq. (A19) in Appendix)

$$\begin{cases} c_{d2} = -\frac{2\Delta c_{d1} + 1}{2\Delta + 1} \\ c_{d3} = \frac{c_{d1} + 2\Delta}{2\Delta + 1} \\ c_{d4} = \frac{-c_{d1} + 1}{2\Delta + 1} \\ c_{d1} \neq 1 \end{cases}. \quad (33)$$

Furthermore, in order to minimize numerical instability one may intuitively require

$$|c_{di}| \leq 1, (i = 1, 2, \text{ and } 3), \quad (34a)$$

so that the following constraint may be desired

$$-1 \leq c_{d1} < 1. \quad (34b)$$

A stability study with a numerical example in Section 3.1.1 shows that (34b) would always guarantee numerical stability for reasonable parameters used. It is also shown that while the upper bound for the numerical stability region is indeed $c_{d1} = 1$, the lower bound can be extended from -1 to certain values depending on the relaxation coefficient τ_{ij} and the lattice fraction Δ .

Note that further effort can be devoted to construct more constraints on the parameter c_{d1} by using more distributions on extra lattice nodes in the interpolation, such as the local third-order Dirichlet schemes provided in [48]. However, considering the LBE is second-order accurate in the interior, we aim at only obtaining a robust second-order boundary treatment scheme in this work. Furthermore, the choice of the relaxation coefficients τ_{ij} will affect the error terms in the TLBE solution. A comprehensive study on the effects of the relaxation coefficients and optimization of the parameters in Eq. (32) is thus

beyond the scope of this work. To demonstrate the applicability and the second-order accuracy of the present boundary condition treatment, we provide the following three particular choices for the Dirichlet condition treatment:

Scheme 1:

$$g_{\bar{z}}(\mathbf{x}_f, t + \delta t) = \begin{cases} (-2\Delta)\hat{g}_z(\mathbf{x}_f, t) + (2\Delta - 1)\hat{g}_z(\mathbf{x}_{ff}, t) + \varepsilon_D\Phi_d, & (0 \leq \Delta \leq 0.5) \\ (-\frac{1}{2\Delta})\hat{g}_z(\mathbf{x}_f, t) + (1 - \frac{1}{2\Delta})\hat{g}_z(\mathbf{x}_f, t) + (\frac{1}{2\Delta})\varepsilon_D\Phi_d, & (\Delta > 0.5) \end{cases} \quad (35a)$$

Scheme 2:

$$g_{\bar{z}}(\mathbf{x}_f, t + \delta t) = 2(\Delta - 1)\hat{g}_z(\mathbf{x}_f, t) - \left(\frac{(2\Delta - 1)^2}{2\Delta + 1}\right)\hat{g}_z(\mathbf{x}_{ff}, t) + 2\left(\frac{2\Delta - 1}{2\Delta + 1}\right)\hat{g}_z(\mathbf{x}_f, t) + \left(\frac{3 - 2\Delta}{2\Delta + 1}\right)\varepsilon_D\Phi_d, \text{ and} \quad (35b)$$

Scheme 3:

$$g_{\bar{z}}(\mathbf{x}_f, t + \delta t) = -\hat{g}_z(\mathbf{x}_f, t) + \left(\frac{2\Delta - 1}{2\Delta + 1}\right)\hat{g}_z(\mathbf{x}_{ff}, t) + \left(\frac{2\Delta - 1}{2\Delta + 1}\right)\hat{g}_z(\mathbf{x}_f, t) + \left(\frac{2}{2\Delta + 1}\right)\varepsilon_D\Phi_d. \quad (35c)$$

The three schemes in Eq. (35) all have second-order accuracy in space and they reduce to the ordinary bounce-back scheme in Eq. (30) when $\Delta = 0.5$. They are all computationally stable.

One may also observe that Scheme 1 is very similar to the hydrodynamic boundary condition treatment by Bouzidi et al. [39] which is based on the bounce-back rule and linear interpolation. The present analysis and the derivation in Appendix can also offer some insights into the understanding of the various interpolation schemes for velocity boundary conditions as those in [32,33,36–39]. We notice that, however, a rigorous second-order accurate scheme for the Dirichlet velocity boundary condition cannot be similarly formulated, even though for some special cases such as pressure-driven channel flows second-order accuracy of the velocity field can be obtained for arbitrary Δ with the coefficients carefully chosen based on interpolation. The reason is that in athermal LBE simulations, in principle, the leading-order and first-order moments of the velocity distribution functions must be able to recover the mass and momentum equations, respectively, which necessitates more constraints on the velocity distribution functions. Some of the relationships for the temperature distribution functions in TLBE do not have their corresponding forms for the velocity distribution functions in athermal LBE simulations.

Ginzburg [48] showed that with five populations incorporated, a “third-order” Dirichlet condition treatment can be constructed with one adjustable coefficient. With the present notation, the “one-parameter family of third-order conditions” in [48] becomes

$$g_{\bar{z}}(\mathbf{x}_f, t + \delta t) = \frac{1}{\kappa + 1} \left[\left(\frac{1}{2} - \Delta - \frac{1}{2}\Delta^2 + \kappa \right) \hat{g}_z(\mathbf{x}_f, t) + \left(-\frac{3}{2} + \frac{3}{2}\Delta + \Delta^2 - 2\kappa \right) \hat{g}_z(\mathbf{x}_{ff}, t) + \left(\frac{1}{2} - \frac{1}{2}\Delta - \frac{1}{2}\Delta^2 + \kappa \right) \hat{g}_z(\mathbf{x}_{fff}, t) + \left(1 - \frac{1}{2}\Delta + 2\kappa \right) \hat{g}_z(\mathbf{x}_f, t) + \left(-\frac{1}{2} + \frac{1}{2}\Delta - \kappa \right) \hat{g}_z(\mathbf{x}_{ff}, t) + \frac{1}{2}\varepsilon_D\Phi_d \right], \quad (36)$$

where $\mathbf{x}_{fff} = \mathbf{x}_{ff} + \mathbf{e}_{\bar{z}}$, $\delta t = \mathbf{x}_f + 2\mathbf{e}_{\bar{z}}$, δt and κ is an adjustable coefficient. Two particular solutions were provided in [48] by choosing $\kappa = 0$ and $\kappa = (-1 + 2\Delta + \Delta^2)/2$ in Eq. (36). In addition, the particular second-order Dirichlet condition applied in [48] used the same three populations as in the present treatment and can be obtained with $c_{d1} = (1/2 - \Delta)$ in Eq. (33), which yields

$$g_{\bar{z}}(\mathbf{x}_f, t + \delta t) = \left(\frac{1}{2} - \Delta \right) \hat{g}_z(\mathbf{x}_f, t) + (\Delta - 1)\hat{g}_z(\mathbf{x}_{ff}, t) + \frac{1}{2}\hat{g}_z(\mathbf{x}_f, t) + \frac{1}{2}\varepsilon_D\Phi_d. \quad (37)$$

2.3.2. Neumann boundary condition

In order to develop an appropriate boundary condition for a curved wall with an arbitrary Δ and arbitrary orientation, we first consider the case when the wall normal is aligned with $\mathbf{e}_{\bar{z}}$ and $\mathbf{e}_z$ as in (31). Hence, the boundary condition for $g_{\bar{z}}(\mathbf{x}_f, t + \delta t)$ can be represented as

$$g_{\bar{z}}(\mathbf{x}_f, t + \delta t) = c_{n1}\hat{g}_z(\mathbf{x}_f, t) + c_{n2}\hat{g}_z(\mathbf{x}_{ff}, t) + c_{n3}\hat{g}_z(\mathbf{x}_f, t) + c_{n4}(\delta t/\delta x)\Phi_{n\bar{z}}, \quad (38)$$

where c_{n1} , c_{n2} , c_{n3} , and c_{n4} are coefficients related to Δ , and $\Phi_{n\bar{z}} = \Phi_n$.

Different from the adjustability of the treatment for the Dirichlet condition, the second-order accurate scheme for the Neumann boundary condition is determined by (see Eq. (A28) in Appendix)

$$\begin{cases} c_{n1} = 1 \\ c_{n2} = -\frac{2\Delta-1}{2\Delta+1} \\ c_{n3} = \frac{2\Delta-1}{2\Delta+1} \\ c_{n4} = \frac{2}{2\Delta+1} \end{cases}. \quad (39)$$

Thus Eq. (38) can be explicitly expressed as

$$g_x(\mathbf{x}_f, t + \delta t) = \hat{g}_x(\mathbf{x}_f, t) - \left(\frac{2\Delta - 1}{2\Delta + 1} \right) \hat{g}_x(\mathbf{x}_{ff}, t) + \left(\frac{2\Delta - 1}{2\Delta + 1} \right) \hat{g}_x(\mathbf{x}_f, t) + \left(\frac{2}{2\Delta + 1} \right) \frac{\delta t}{\delta x} \Phi_{nz}. \quad (40)$$

Eq. (40) reduces to the ordinary bounce-back scheme in Eq. (31) when $\Delta = 0.5$ and its coefficients satisfy the stability requirement $|c_{ni}| \leq 1$ ($i = 1, 2$, and 3).

Remark 1. The present Dirichlet and Neumann boundary condition treatments given by Eqs. (32) and (38) with their coefficients determined by Eqs. (33) and (39), respectively, are derived for straight boundary surfaces with arbitrary Δ values and with their normal direction parallel to the discrete velocity direction $\mathbf{e}_z$. For curved boundaries, Eq. (32) can be directly extended for Dirichlet problems and Eq. (38) constitutes an excellent starting point for general Neumann problems.

Remark 2. Eq. (38) is also valid for Dirichlet problems, and Eq. (32) is also applicable to Neumann problems. Thus the combination of Eqs. (32) and (38) gives a means to evaluate the heat flux on the wall along $\mathbf{e}_z$ direction for Dirichlet problems and to evaluate the wall temperature at the cuts for Neumann problems. With the coefficients given by Eqs. (33) and (39), the following expressions can be obtained on the wall intersected by the lattice vector $\mathbf{e}_z$

$$\Phi_{nz} = \left(\frac{1 - c_{d1}}{2} \right) \frac{\delta x}{\delta t} \left[-(2\Delta + 1) \hat{g}_x(\mathbf{x}_f, t) + \left(2\Delta - \frac{2}{1 - c_{d1}} \right) \hat{g}_x(\mathbf{x}_{ff}, t) + \left(\frac{2}{1 - c_{d1}} - 1 \right) \hat{g}_x(\mathbf{x}_f, t) + \varepsilon_D \Phi_d \right], \quad (41a)$$

or

$$\Phi_d = \frac{1}{\varepsilon_D} \left[(2\Delta + 1) \hat{g}_x(\mathbf{x}_f, t) + \left(\frac{2}{1 - c_{d1}} - 2\Delta \right) \hat{g}_x(\mathbf{x}_{ff}, t) + \left(1 - \frac{2}{1 - c_{d1}} \right) \hat{g}_x(\mathbf{x}_f, t) + \left(\frac{2}{1 - c_{d1}} \right) \frac{\delta t}{\delta x} \Phi_{nz} \right], \quad (41b)$$

where c_{d1} is the aforementioned adjustable variable in Eq. (32) and its values based on the three different choices in Eq. (35a–c) will be examined subsequently.

Remark 3. The temperature gradient within the interior of the field can be evaluated using the formula discussed in [28]

$$\tau_{ij} \frac{\partial \phi}{\partial x_j} = \frac{v_i \delta t}{\varepsilon_D (\delta x)^2} \phi - \frac{1}{\varepsilon_D \delta x} \sum_{\alpha=0}^m e_{\alpha i} g_{\alpha}. \quad (42)$$

This allows for an assessment of the error propagation in terms of heat flux into the field due to the boundary condition treatment.

Remark 4. For Neumann problems with curved boundaries, the heat flux Φ_{nz} in the discrete velocity direction must be obtained based on the given Φ_n in the absence of explicit knowledge on the tangential derivative, Φ_t , along the wall for curved boundaries (see Fig. 3). A practical strategy for the numerical determination of Φ_{nz} is discussed next.

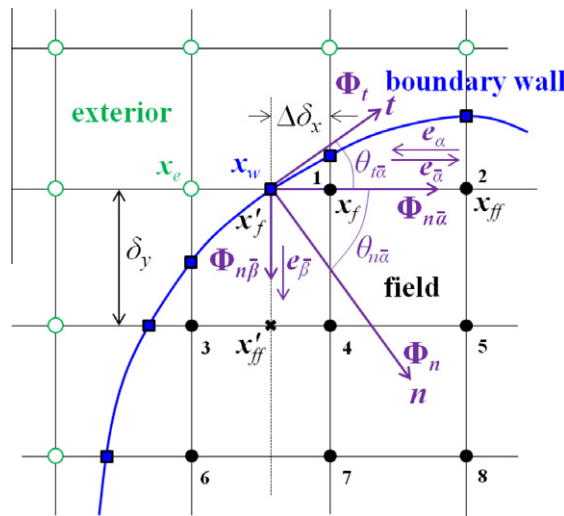


Fig. 3. Layout of the rectangular lattice and a curved boundary (black circles: field nodes; blue squares: boundary nodes; white circles: exterior nodes). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Consider a curved boundary wall intersected by the lattice as shown in Fig. 3. The wall cuts the horizontal lattice link along $\mathbf{e}_x$ direction between $\mathbf{x}_e$ and $\mathbf{x}_f$ at $\mathbf{x}_w$, and the heat flux along the link direction ($\mathbf{e}_x$) from exterior to interior, $\Phi_{n\bar{x}}$, is to be derived. The angles between $\mathbf{e}_x$ and the wall tangent and normal are θ_{tx} and $\theta_{n\bar{x}}$, respectively. Along $\mathbf{e}_x$, the two neighboring field lattice nodes are $\mathbf{x}_f$ and $\mathbf{x}_{ff}$; along the direction perpendicular to $\mathbf{e}_x$, we designate $\mathbf{x}'_f = \mathbf{x}_w$ and $\mathbf{x}'_{ff} = \mathbf{x}_w + \mathbf{e}_\beta \delta t$ in which $\mathbf{e}_\beta$ is the lattice vector from exterior to the interior along the direction perpendicular to $\mathbf{e}_x$.

The projection of the boundary fluxes gives $\Phi_{n\bar{x}} = \Phi_n \cos \theta_{n\bar{x}} + \Phi_t \cos \theta_{tx}$. In general, Φ_t is needed in the calculation of $\Phi_{n\bar{x}}$, unless $\cos \theta_{tx} = 0$ (i.e., the wall is perpendicular to $\mathbf{e}_x$) or $\Phi_t = 0$ is implicitly known in very special cases. However, well-posedness of the mathematical problem for CDE does not even allow the specification of Φ_t in addition to Φ_n on the same boundary. Thus the Neumann boundary condition, or the more general mixed boundary condition in the form of

$$\sigma_1 \frac{\partial \phi}{\partial n} + \sigma_2 \phi = \sigma_3, (\sigma_1 \neq 0) \quad (43)$$

involving the normal derivative on the boundary requires a fundamentally different approach than the Dirichlet problem. By rearranging Eq. (43), the boundary heat flux in the normal direction can be written as

$$\Phi_n = -D_t \frac{\partial \phi}{\partial n} = -D_t \left(\frac{\sigma_3}{\sigma_1} - \frac{\sigma_2}{\sigma_1} \phi \right) = -D_t \left(\frac{\sigma_3}{\sigma_1} - \frac{\sigma_2}{\sigma_1} \Phi_d \right). \quad (44)$$

where D_t is the thermal diffusion coefficient on the boundary and the standard Neumann condition is recovered by setting $\sigma_2 = 0$ in the above.

For brevity, only the D2Q5 TLBE model in 2-D is considered here as the extension to 3-D is straightforward. As in Fig. 3 the normal heat flux can be expressed as

$$\Phi_{n\bar{x}} \cos \theta_{n\bar{x}} + \Phi_{n\bar{\beta}} \cos \theta_{n\bar{\beta}} = \Phi_n, \quad (45)$$

where $\theta_{n\bar{\beta}}$ is the angle between $\mathbf{e}_\beta$ and the normal $\mathbf{n}$. Additionally, we can eliminate $\mathbf{g}_x(\mathbf{x}_f, t + \delta t)$ by combining Eqs. (32) and (38) to obtain a relation between Φ_d and the heat flux $\Phi_{n\bar{x}}$ in the discrete velocity direction $\mathbf{e}_x$ on the boundary,

$$c_{d4} \varepsilon_D \Phi_d = (c_{n1} - c_{d1}) \hat{g}_x(\mathbf{x}_f, t) + (c_{n2} - c_{d2}) \hat{g}_x(\mathbf{x}_{ff}, t) + (c_{n3} - c_{d3}) \hat{g}_x(\mathbf{x}_f, t) + c_{n4} (\delta t / \delta x) \Phi_{n\bar{x}}. \quad (46a)$$

Eq. (46a) is actually an alternative form of Eq. (41). It is important to recognize that a similar relation can be derived between Φ_d and the heat flux $\Phi_{n\bar{\beta}}$ in $\mathbf{e}_\beta$ direction as well. That is

$$c'_{d4} \varepsilon_D \Phi_d = (c'_{n1} - c'_{d1}) \hat{g}_\beta(\mathbf{x}'_f, t) + (c'_{n2} - c'_{d2}) \hat{g}_\beta(\mathbf{x}'_{ff}, t) + (c'_{n3} - c'_{d3}) \hat{g}_\beta(\mathbf{x}'_f, t) + c'_{n4} (\delta t / \delta x) \Phi_{n\bar{\beta}}, \quad (46b)$$

where c'_{di} , c'_{ni} are the coefficients corresponding to c_{di} , c_{ni} ($i = 1, 2, 3, 4$) defined in Eqs. (33) and (39), respectively, with the fraction $\Delta_\beta = 0$ in $\mathbf{e}_\beta$ direction. The distribution functions $\hat{g}_\beta(\mathbf{x}'_f, t)$, $\hat{g}_\beta(\mathbf{x}'_{ff}, t)$, and $\hat{g}_\beta(\mathbf{x}'_f, t)$ at $\mathbf{x}'_f$ and $\mathbf{x}'_{ff}$ can be obtained using a quadratic interpolation or extrapolation of the distributions at the closest interior lattice nodes. Further elimination of Φ_d from Eqs. (46a–b) results in the following relation between $\Phi_{n\bar{x}}$ and $\Phi_{n\bar{\beta}}$,

$$\begin{aligned} & \frac{1}{c_{d4}} [(c_{n1} - c_{d1}) \hat{g}_x(\mathbf{x}_f, t) + (c_{n2} - c_{d2}) \hat{g}_x(\mathbf{x}_{ff}, t) + (c_{n3} - c_{d3}) \hat{g}_x(\mathbf{x}_f, t) + c_{n4} (\delta t / \delta x) \Phi_{n\bar{x}}] \\ &= \frac{1}{c'_{d4}} [(c'_{n1} - c'_{d1}) \hat{g}_\beta(\mathbf{x}'_f, t) + (c'_{n2} - c'_{d2}) \hat{g}_\beta(\mathbf{x}'_{ff}, t) + (c'_{n3} - c'_{d3}) \hat{g}_\beta(\mathbf{x}'_f, t) + c'_{n4} (\delta t / \delta x) \Phi_{n\bar{\beta}}]. \end{aligned} \quad (47)$$

Finally, combining Eqs. (45) and (47) leads to the desired expression for $\Phi_{n\bar{x}}$ in terms of the normal heat flux and the post-collision distributions in the neighboring lattice nodes,

$$\begin{aligned} \Phi_{n\bar{x}} = & \left\{ \frac{1}{c'_{d4}} [(c'_{n1} - c'_{d1}) \hat{g}_\beta(\mathbf{x}'_f, t) + (c'_{n2} - c'_{d2}) \hat{g}_\beta(\mathbf{x}'_{ff}, t) + (c'_{n3} - c'_{d3}) \hat{g}_\beta(\mathbf{x}'_f, t)] \sin \theta_{n\bar{x}} \right. \\ & \left. - \frac{1}{c_{d4}} [(c_{n1} - c_{d1}) \hat{g}_x(\mathbf{x}_f, t) + (c_{n2} - c_{d2}) \hat{g}_x(\mathbf{x}_{ff}, t) + (c_{n3} - c_{d3}) \hat{g}_x(\mathbf{x}_f, t)] \sin \theta_{n\bar{x}} + \frac{c'_{n4}}{c'_{d4}} \frac{\delta t}{\delta x} \Phi_n \right\} \left/ \left[\frac{c_{n4}}{c_{d4}} \frac{\delta t}{\delta x} \sin \theta_{n\bar{x}} + \frac{c'_{n4}}{c'_{d4}} \frac{\delta t}{\delta x} \cos \theta_{n\bar{x}} \right] \right. \end{aligned} \quad (48)$$

Eq. (48) can be directly applied to Neumann problems on curved boundaries. For the special case where $\Phi_t = 0$ or $\cos \theta_{tx} = 0$, a simpler formula can be used to relate $\Phi_{n\bar{x}}$ to Φ_n

$$\Phi_{n\bar{x}} = \Phi_n \cos \theta_{n\bar{x}}. \quad (49)$$

The above is equivalent to that used by Huang et al. [42], but it should not be used in general.

For mixed boundary conditions, Φ_n is not specified. Instead, Φ_n is related to Φ_d through Eqs. (44, 46). Substitution of Eq. (46a) into Eq. (44) yields

$$\Phi_n = -D_t \frac{\sigma_3}{\sigma_1} + D_t \frac{\sigma_2}{\sigma_1 c_{d4} \varepsilon_D} [(c_{n1} - c_{d1}) \hat{g}_x(\mathbf{x}_f, t) + (c_{n2} - c_{d2}) \hat{g}_x(\mathbf{x}_{ff}, t) + (c_{n3} - c_{d3}) \hat{g}_x(\mathbf{x}_f, t) + c_{n4} (\delta t / \delta x) \Phi_{n\bar{x}}]. \quad (50)$$

Combining Eqs. (48) and (50) gives

$$\begin{aligned} \Phi_{nz} = & \left\{ \frac{1}{c'_{d4}} [(c'_{n1} - c'_{d1}) \hat{g}_\beta(\mathbf{x}'_f, t) + (c'_{n2} - c'_{d2}) \hat{g}_\beta(\mathbf{x}'_{ff}, t) + (c'_{n3} - c'_{d3}) \hat{g}_\beta(\mathbf{x}'_f, t)] \sin \theta_{nz} - \frac{1}{c'_{d4}} [(c_{n1} - c_{d1}) \hat{g}_\alpha(\mathbf{x}_f, t) \right. \\ & + (c_{n2} - c_{d2}) \hat{g}_\alpha(\mathbf{x}_{ff}, t) + (c_{n3} - c_{d3}) \hat{g}_\alpha(\mathbf{x}_f, t)] \sin \theta_{nz} + D_t \frac{c'_{n4}}{c'_{d4}} \frac{\delta t}{\delta x} \frac{\sigma_2}{\sigma_1 c_{d4} \varepsilon_D} [(c_{n1} - c_{d1}) \hat{g}_\alpha(\mathbf{x}_f, t) + (c_{n2} - c_{d2}) \hat{g}_\alpha(\mathbf{x}_{ff}, t) \\ & \left. + (c_{n3} - c_{d3}) \hat{g}_\alpha(\mathbf{x}_f, t)] - D_t \frac{c'_{n4}}{c'_{d4}} \frac{\delta t}{\delta x} \frac{\sigma_3}{\sigma_1} \right\} / \left[\frac{c_{n4}}{c_{d4}} \frac{\delta t}{\delta x} \sin \theta_{nz} + \frac{c'_{n4}}{c'_{d4}} \frac{\delta t}{\delta x} \cos \theta_{nz} - \frac{c'_{n4}}{c'_{d4}} \left(\frac{\delta t}{\delta x} \right)^2 D_t \frac{c_{n4} \sigma_2}{c_{d4} \varepsilon_D \sigma_1} \right]. \end{aligned} \quad (51)$$

Once Φ_{nz} is obtained the boundary condition treatment in Eq. (40) can be used to complete the streaming step. It is noted that $\Phi_{n\bar{\beta}}$ is not needed in Eq. (40) as $\mathbf{x}_w$ in Fig. 3 is located on the link along $\mathbf{e}_\alpha$ direction. The same procedure can be applied to derive $\Phi_{n\bar{\beta}}$ when $\mathbf{x}_w$ is located on the link along $\mathbf{e}_\beta$ direction.

The accuracy of the present boundary condition treatments will be examined through a series of numerical tests in the next section, and the asymptotic analysis for the order-of-accuracy is given in Appendix. When the node $\mathbf{x}_{ff}$ is outside the field, we recommend using the localized schemes by Yoshida and Nagaoka [28].

3. Results and discussion

To verify the applicability and accuracy of the present thermal boundary condition treatments, we have conducted several numerical tests involving both 2-D and 3-D geometries. Both convection and diffusion are considered in the first example, where the Dirichlet and Neumann boundary conditions are separately applied and investigated. The second example shows how to convert the Neumann condition in the normal direction of an inclined wall into the heat fluxes in the discrete velocity directions when there is no tangential flux. From the third example through the fifth example, curved geometries are considered. Steady-state and transient boundary conditions are given in the third and fourth problems; while the fifth test is for the simulation of steady-state convection and diffusion inside a circular (3-D) pipe.

In this study, only isotropic diffusion is considered, i.e., $D_{ij} = D_t \delta_{ij}$; thus the off-diagonal components in the relaxation matrix are set to be zero, and the constraint for the diagonal coefficients is maintained so that Eq. (14) can be rewritten as

$$\tau_{ij} = \tau_D \delta_{ij} = \left(\frac{1}{2} + \frac{\delta t}{\varepsilon_D (\delta x)^2} D_t \right) \delta_{ij}. \quad (52)$$

The positivity of the diffusion coefficient D_t requires that $\tau_D > 1/2$. A typical value of $\tau_D = 0.75$ is used in each of the numerical examples to demonstrate the accuracy of the present boundary condition treatments. Although no theoretical analysis on the numerical stability of the proposed Dirichlet and Neumann boundary schemes is given, we have investigated their stability by choosing τ_D that is close to 0.5 when numerical instability is most likely to occur. For brevity, the results at $\tau_D = 0.55$ and $\tau_D = 0.51$ are presented for numerous cases. Furthermore, a detailed stability study is carried out in Section 3.1.1 to find the stability bounds for the 2-D channel flow with a Dirichlet condition; and all three schemes given by Eq. (35a–c) are well within the stability boundaries. The relaxation coefficient τ_0 for the conserved quantity ϕ has no effect on the numerical solution, and the remaining adjustable coefficients do not contribute to the leading-order solution of the CDE since they only affect the higher-order error terms of the LBE solutions; thus $\tau_0 = 0$, and $\tau_p = 1$ ($p = 4, 5, 6$ in 3-D and $p = 3, 4$ in 2-D) are used in the rest of this work.

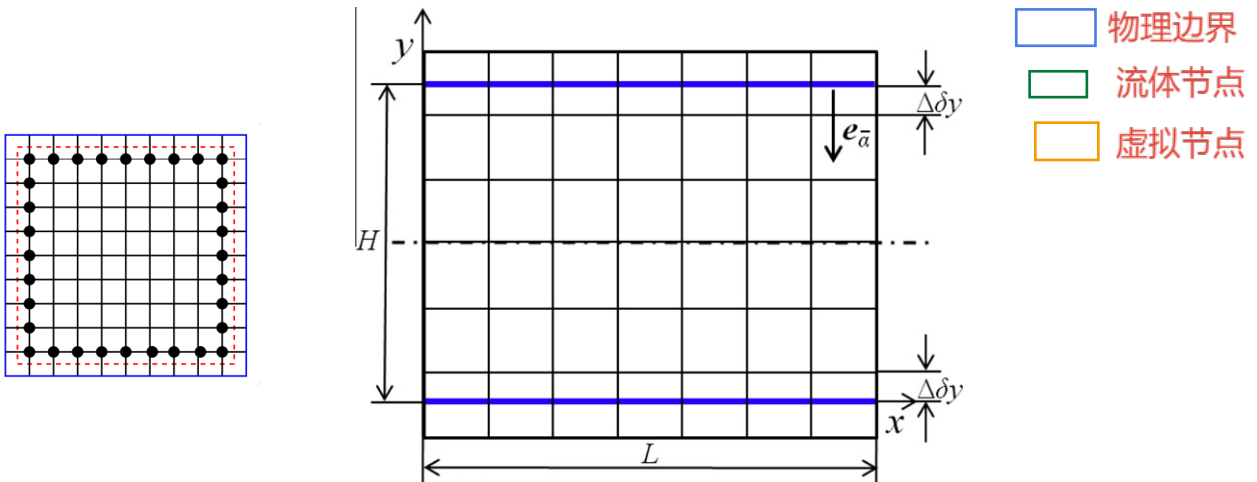


Fig. 4. Schematic representation of the lattice in a 2-D channel with an arbitrary Δ value.

3.1. Two-D steady-state channel flow

First we consider a plug flow of velocity U in a 2-D channel of constant height, H (Fig. 4). Each wall is placed at a distance of $(\Delta\delta y)$ from the first interior lattice in the y -direction. The CDE for the temperature inside the channel is

$$U \frac{\partial T}{\partial x} = D_t \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right), \quad (53)$$

This thermal flow problem is characterized by the Péclet number $Pe = HU/D_t$, and $Pe = 20$ is used in all the simulations. Periodic boundary conditions are imposed in the x -direction so that $f(x+L) = f(x)$ for both the temperature T and distribution functions g_x . Both Dirichlet and Neumann thermal boundary conditions on the walls are considered.

3.1.1. Dirichlet boundary condition

On the walls at $y = 0$ and $y = H$,

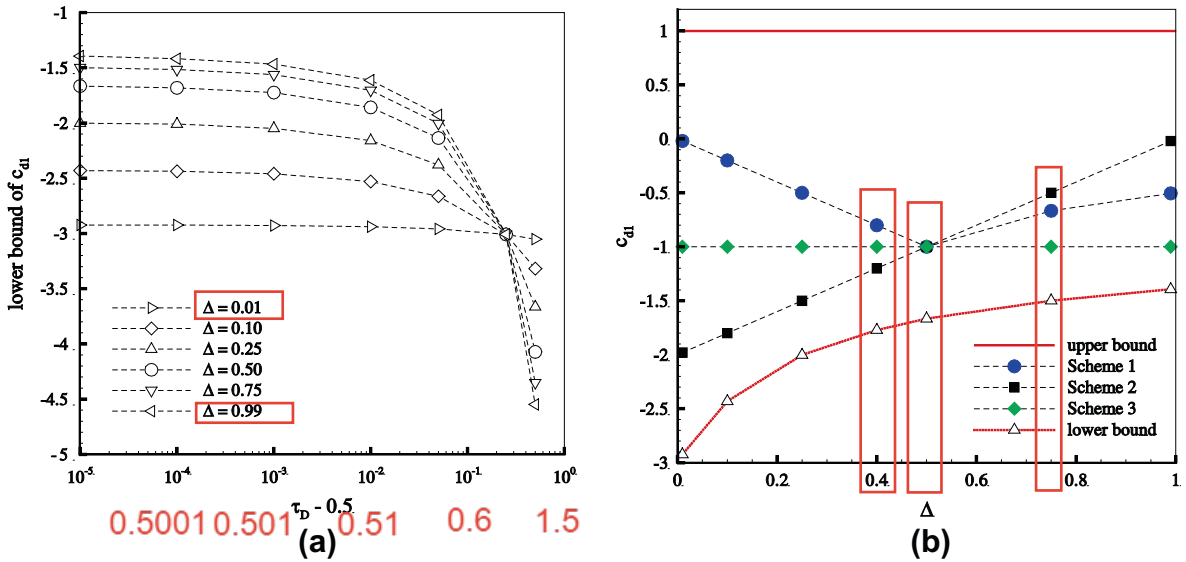


Fig. 5. (a) Lower stability bound of the adjustable parameter c_{d1} versus $(\tau_D - 0.5)$, and (b) variations of c_{d1} as a function of the lattice fraction, Δ , for the present Dirichlet schemes at $\tau_D = 0.50001$ for the channel flow Dirichlet problem (the lower and upper stability bounds are included).

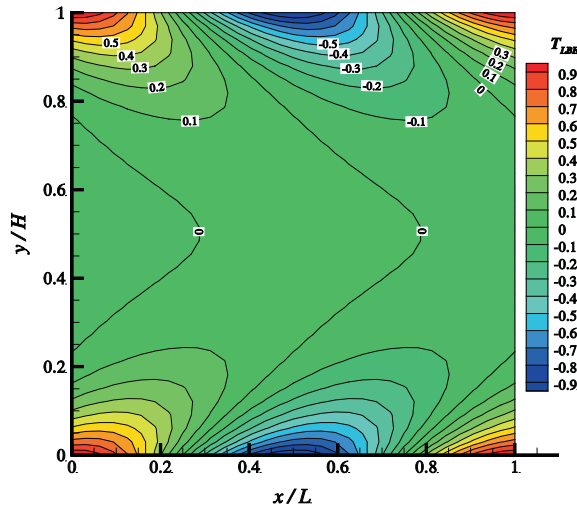


Fig. 6. Temperature contours in a 2-D channel with Dirichlet boundary condition using $\tau_D = 0.75$, $\Delta = 0.5$ and $H = 64$. All three schemes reduce to the “bounce-back” condition.

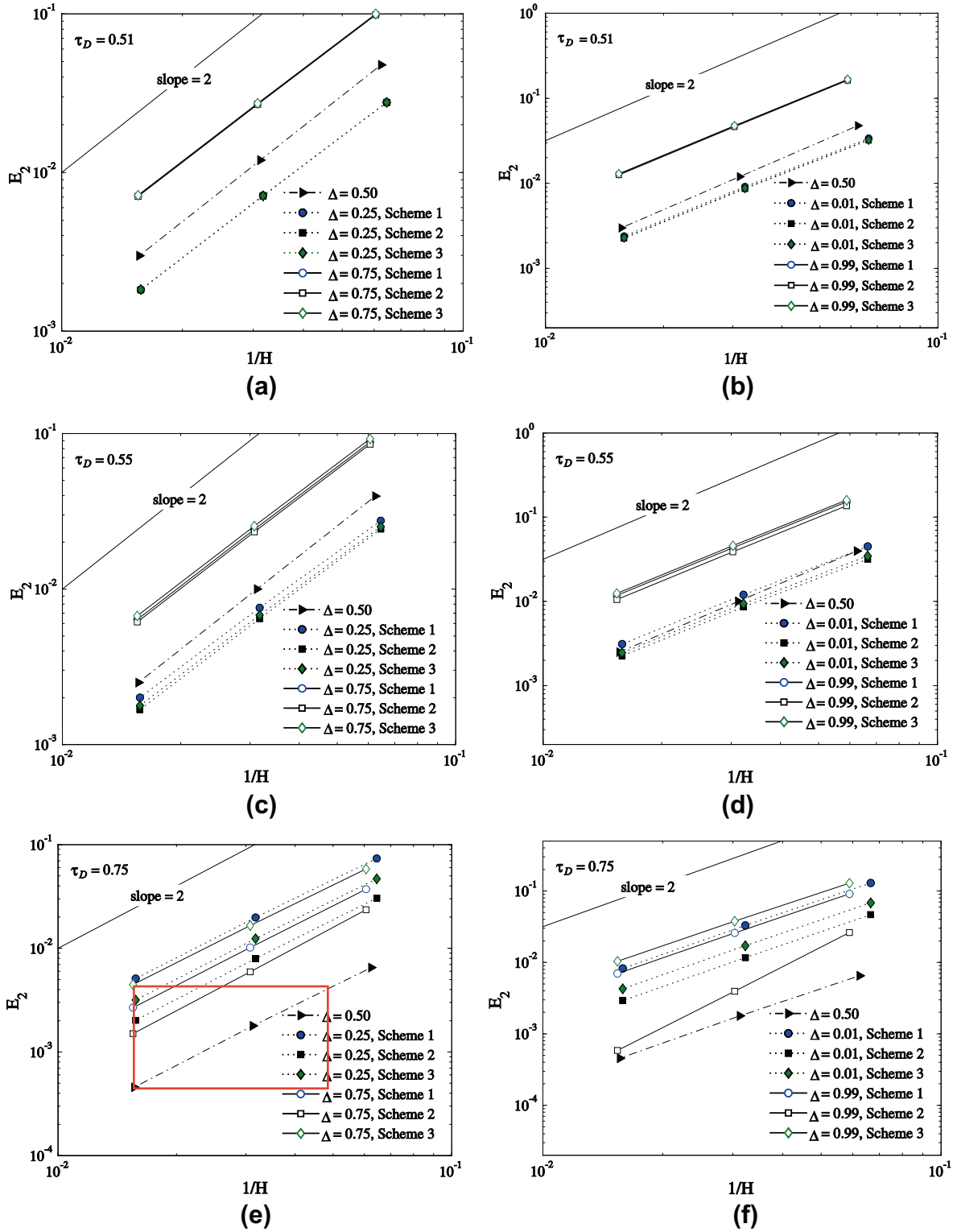


Fig. 7. Relative L_2 error norm E_2 versus the grid resolution, $1/H$, for the channel flow Dirichlet problem. (a) $\Delta = 0.50, 0.25$, and 0.75 at $\tau_D = 0.51$, (b) $\Delta = 0.50, 0.01$, and 0.99 at $\tau_D = 0.51$, (c) $\Delta = 0.50, 0.25$, and 0.75 at $\tau_D = 0.55$, (d) $\Delta = 0.50, 0.01$, and 0.99 at $\tau_D = 0.55$, (e) $\Delta = 0.50, 0.25$, and 0.75 at $\tau_D = 0.75$, (f) $\Delta = 0.50, 0.01$, and 0.99 at $\tau_D = 0.75$.

$$T(x, y = 0) = T(x, y = H) = \cos(kx), \text{ with } k = 2\pi/L. \quad (54)$$

An analytical solution for the thermal field can be obtained. When expressed with complex variables, the exact solution is

$$T_{ex}(x, y) = \text{Real} \left[e^{ikx} \left(\frac{1 - e^{-\lambda H}}{e^{\lambda H} - e^{-\lambda H}} e^{\lambda y} - \frac{1 - e^{\lambda H}}{e^{\lambda H} - e^{-\lambda H}} e^{-\lambda y} \right) \right], \quad (55)$$

where “Real” means taking the real part of a complex variable and $\lambda = k\sqrt{1 + \frac{iU}{Dk}}$.

Before evaluating the accuracy of the present Dirichlet schemes, the numerical stability is investigated first for this problem. The work of [37] indicates that the numerical instability associated with boundary condition treatments in using the LBE method is more pronounced in channel flows than in curved geometries. Hence guidance on numerical instability from solving the thermal problem in the channel flow will be useful. Fig. 5(a) shows the lower stability bound for the adjustable coefficient c_{d1} at $Pe = 20$ and $Ny = 34$ (Ny denotes the number of lattice nodes in y -direction). Very close results for the lower bound are obtained also for $(Pe, Ny) = (20, 130)$, $(100, 34)$ and $(100, 130)$. The upper bound for c_{d1} is 1.0 for all cases, noting that $c_{d1} \neq 1$ as required in Eq. (33). An asymptotic lower bound of c_{d1} is noticed for each Δ value tested as τ_D approaches 0.5 in Fig. 5(a). On the other hand, when τ_D increases, the lower bound drops, resulting in a broader stable region while the overall relative error of the temperature field increases (see Figs. 7 and 8 below). Fig. 5(b) shows the dependence of c_{d1} on Δ in the range of $0 \leq \Delta < 1$ for the three schemes in relation to the lower stability bound for $\tau_D = 0.50001$ and the upper stability bound. Clearly, all three schemes are well within the stable region.

As an illustration, Fig. 6 shows the temperature contours obtained using $\tau_D = 0.75$, $\Delta = 0.5$ and $H = 64$. To assess the accuracy of the LBE solution of the temperature T_{LBE} the following **relative L-2 error norm** is defined:

$$E_2 = \left(\frac{\sum_{x,y} (T_{LBE} - T_{ex})^2}{\sum_{x,y} T_{ex}^2} \right)^{1/2}. \quad (56)$$

Fig. 7(a–f) shows the **log-log plots of E_2 versus the dimensionless grid resolution, $1/H$** , over a complete range of Δ for the three schemes given in Eq. (35a–c). Three relaxation coefficient values $\tau_D = 0.51, 0.55$ and 0.75 are used. **Second-order accuracy is observed for all cases**. At $\tau_D = 0.51$, all three schemes give very close results as shown in Fig. 7(a and b). As τ_D increases, the differences among three schemes become more noticeable and **Scheme 2 gives smallest relative error for each case** as shown in Fig. 7(c–f). The variations of E_2 in the range of Δ from 0 to 1 are shown in Fig. 8 with $Ny = 34$. It is noticed that with $\tau_D = 0.75$, all the three schemes have minimum error at $\Delta = 0.5$, where the standard thermal bounce-back scheme is applied. When τ_D increases to 1.0, the relative error also increases for each scheme when $\Delta \leq 0.5$; while for $\Delta > 0.5$, the error behavior becomes more complicated. The results in Figs. 7 and 8 clearly verify the existence of the adjustable parameter c_{d1} in using the TLBE method for CDE subjected to Dirichlet boundary conditions; and the proposed Dirichlet schemes are stable and of second-order accuracy for $\tau_D > 0.5$.

For comparison purposes, the results of E_2 obtained using the “third-order” and second-order schemes in [48] for $\Delta = 0.25, 0.50$ and 0.75 are compared with that from Scheme 2 in Fig. 9(a and b), where the “Ginzburg 3rd-order # 1” and “Ginzburg 3rd-order # 2” represent the two particular schemes whose coefficients are constructed with $\kappa = 0$ and $\kappa = (-1 + 2\Delta + \Delta^2)/2$ in Eq. (36), respectively, and the coefficients for the second-order scheme are given in Eq. (37). Fig. 9 demonstrates that the “third-order” Dirichlet schemes in [48] also lead to second-order convergence since the TLBE method is second-order accurate in the interior. In addition, Scheme 2 gives smaller error than the local “third-order schemes in [48] for each Δ value tested in Fig. 9(a and b).

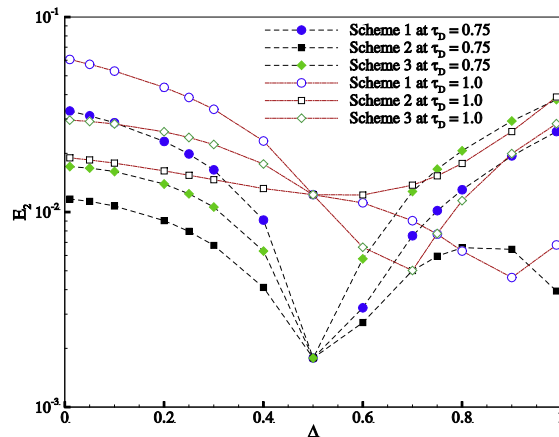


Fig. 8. Relative L-2 error norm E_2 versus the boundary lattice fraction, Δ , for the channel flow Dirichlet problem at $Ny = 34$.

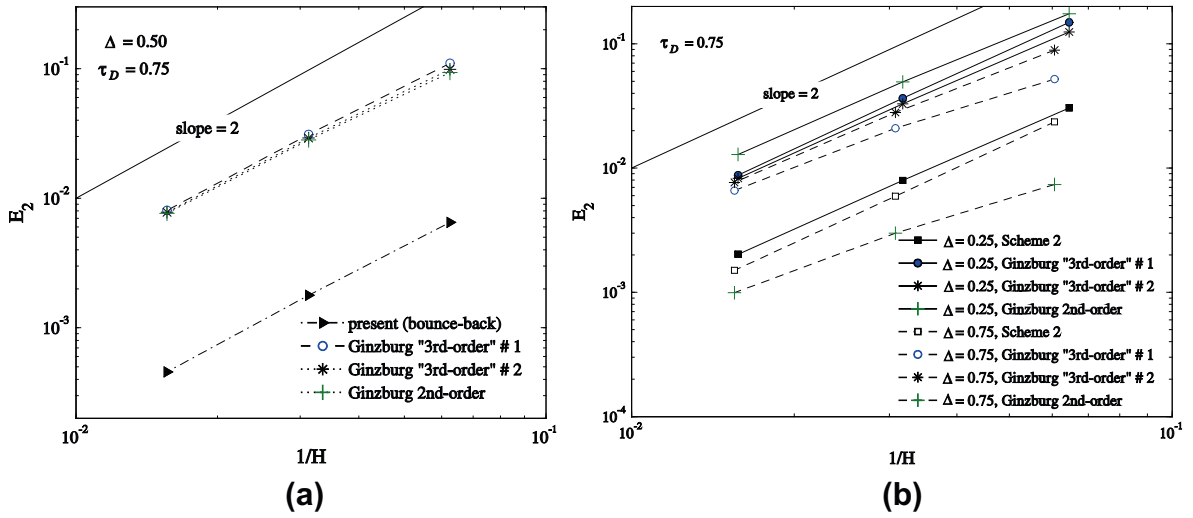


Fig. 9. Comparison of E_2 obtained using Scheme 2 with those using Ginzburg's schemes [48] for the channel flow Dirichlet problem at $\tau_D = 0.75$. (a) $\Delta = 0.50$ (all three schemes reduce to the "bounce-back" scheme), (b) $\Delta = 0.25$ and 0.75 .

To gain further insight into the impact of boundary condition treatments on the accuracy of TLBE solutions, we compute the boundary heat flux and interior temperature gradient using the formulas given by Eqs. (41a) and (42), respectively. The relative L_2 error norms are computed using the respective derivatives obtained from the analytical solutions given by Eq. (55):

$$E_{2_qw} = \left[\sum_{\text{boundary nodes}} \left(\left(\frac{\partial T}{\partial y} \right)_{\text{LBE}} - \left(\frac{\partial T}{\partial y} \right)_{\text{ex}} \right)^2 / \sum_{\text{boundary nodes}} \left(\frac{\partial T}{\partial y} \right)_{\text{ex}}^2 \right]^{1/2}, \quad (57)$$

$$E_{2_grad} = \left[\sum_{\text{interior nodes}} \left(\left(\frac{\partial T}{\partial y} \right)_{\text{LBE}} - \left(\frac{\partial T}{\partial y} \right)_{\text{ex}} \right)^2 / \sum_{\text{interior nodes}} \left(\frac{\partial T}{\partial y} \right)_{\text{ex}}^2 \right]^{1/2}. \quad (58)$$

The results of E_{2_qw} for the wall heat flux and E_{2_grad} for the interior temperature gradient at $\tau_D = 0.75$ are shown in Figs. 10 and 11, respectively.

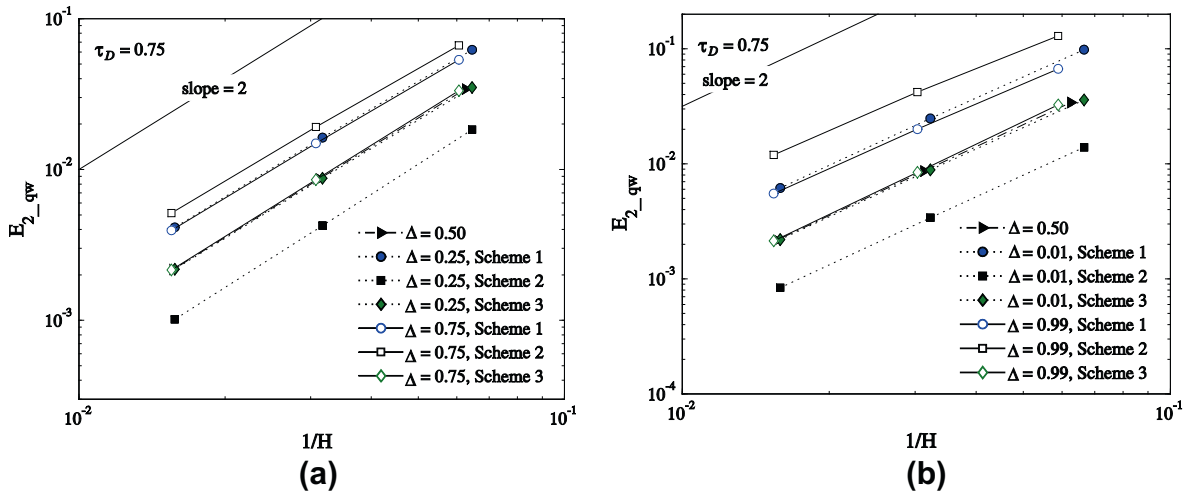


Fig. 10. Wall heat flux error E_{2_qw} versus the grid resolution, $1/H$, for the channel flow Dirichlet problem. (a) $\Delta = 0.50, 0.25$, and 0.75 at $\tau_D = 0.75$, (b) $\Delta = 0.50, 0.01$, and 0.99 at $\tau_D = 0.75$.

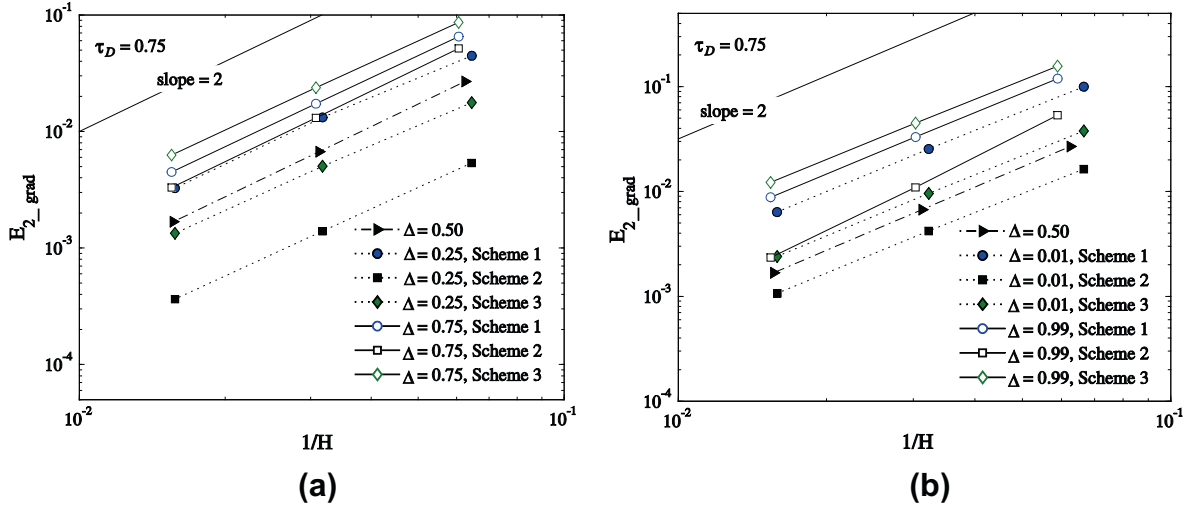


Fig. 11. Interior gradient error $E_{2,\text{grad}}$ versus the grid resolution, $1/H$, for the channel flow Dirichlet problem. (a) $\Delta = 0.50, 0.25$, and 0.75 at $\tau_D = 0.75$, (b) $\Delta = 0.50, 0.01$, and 0.99 at $\tau_D = 0.75$.

It is clear that the numerical solutions for boundary heat flux and interior temperature gradient are of second-order accuracy for Dirichlet problems with straight boundary walls perpendicular to one of the lattice vectors. Scheme 2 gives slightly smaller error than the other two.

3.1.2. Neumann boundary condition

For the Neumann problem, the normal heat flux on the wall is given as

$$q_w = -D_t \left. \frac{\partial T}{\partial y} \right|_{y=H} = -D_t \cos(kx)/H. \quad (59)$$

The exact solution for the temperature is

$$T_{\text{ex}}(x, y) = \text{Real} \left\{ e^{ikx} \left[\frac{e^{-\lambda y} + 1}{\lambda H (e^{\lambda H} - e^{-\lambda H})} e^{\lambda y} + \frac{e^{\lambda H} + 1}{\lambda H (e^{\lambda H} - e^{-\lambda H})} e^{-\lambda y} \right] \right\}. \quad (60)$$

Again, the numerical solution of the temperature field at $\tau_D = 0.75$, $\Delta = 0.5$ and $H = 64$ is shown in Fig. 12 as an example; in addition to the interior temperature and temperature gradient error norms defined in Eqs. (56), (58), the relative L_2 error norm for the boundary temperature is defined as

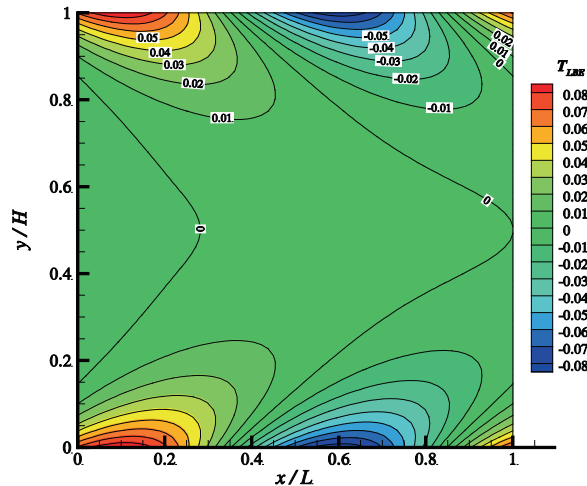


Fig. 12. Temperature contours in a 2-D channel with Neumann boundary condition using $\tau_D = 0.75$, $\Delta = 0.5$ and $H = 64$.

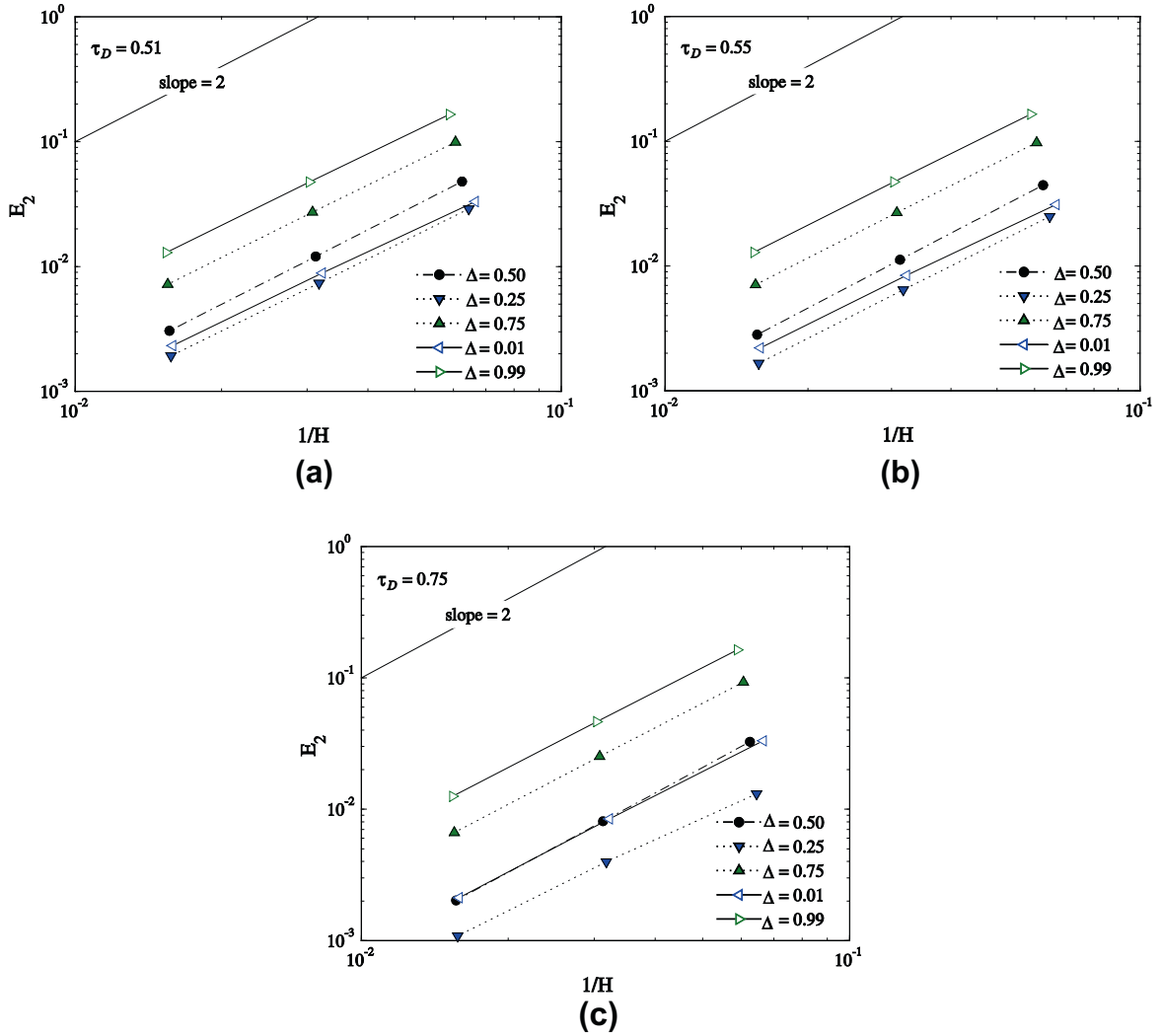


Fig. 13. Relative L_2 error norm E_2 versus the grid resolution, $1/H$, for the channel flow Neumann problem with various Δ . (a) $\tau_D = 0.51$, (b) $\tau_D = 0.55$, and (c) $\tau_D = 0.75$.

$$E_{2_tw} = \left[\frac{\sum_{\text{boundary nodes}} (T_{LBE} - T_{ex})^2}{\sum_{\text{boundary nodes}} T_{ex}^2} \right]^{1/2}. \quad (61)$$

Fig. 13(a–c) show E_2 (Eq. (56)) versus $1/H$ for Δ ranging from 0.01 to 0.99 and $\tau_D = 0.51, 0.55$ and 0.75 at $Pe = 20$, and Figs. 14 and 15 show E_{2_tw} (Eq. (61)) and E_{2_grad} (Eq.(58)), respectively, for $\tau_D = 0.75$. The results confirm that the present Neumann condition scheme is stable, and second-order accuracy is achieved for the temperature field, boundary temperature, and interior temperature gradient when the straight-wall is perpendicular to one of the lattice vectors. Extreme cases including $\tau_D = 0.501, 0.5001, 0.50001$ and $\Delta = 0.001, 0.999, 0.0001, 0.9999$ at $Ny = 34$ have also been examined. The computations using present Neumann boundary scheme are all stable and give good numerical results for all cases.

3.2. One-D transient conduction within an inclined semi-infinite solid with a Neumann condition

In order to demonstrate the applicability of the present Neumann boundary condition treatment to inclined walls, we consider the transient heat conduction into an inclined semi-infinite solid with a constant heat flux supplied at the end as shown in Fig. 16. For a given wall inclination angle θ with respect to the y -axis, the specific fractions of Δ_x and Δ_y are calculated for each field point adjacent to the boundary. Note there is still no tangential variation in this case; the Neumann boundary condition with nonzero tangential gradient will be investigated in Section 3.3.

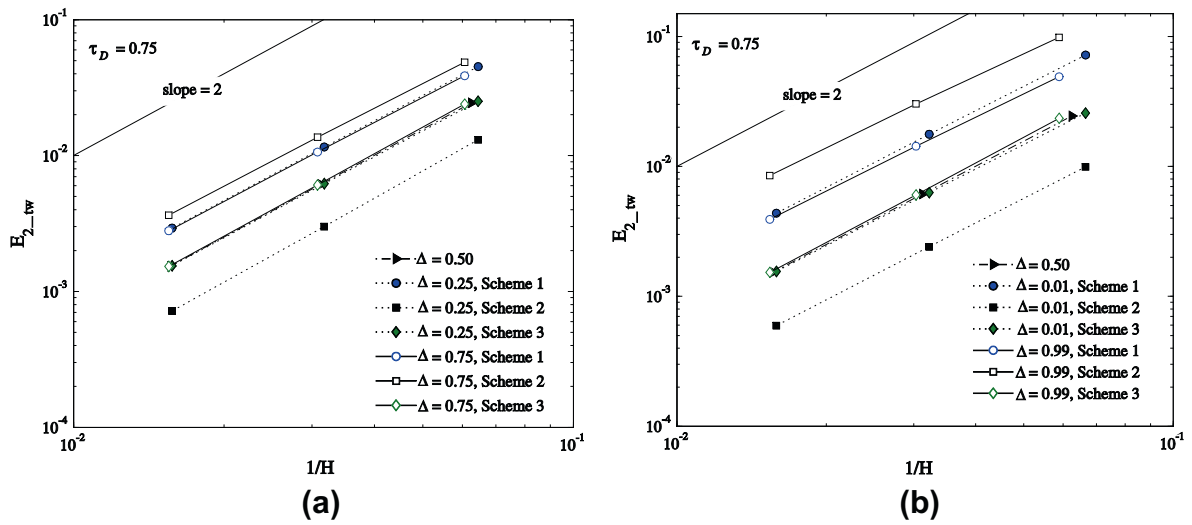


Fig. 14. Wall temperature error $E_{2,tw}$ versus the grid resolution, $1/H$, for the channel flow Neumann problem. (a) $\Delta = 0.50, 0.25$, and 0.75 at $\tau_D = 0.75$, (b) $\Delta = 0.50, 0.01$, and 0.99 at $\tau_D = 0.75$.

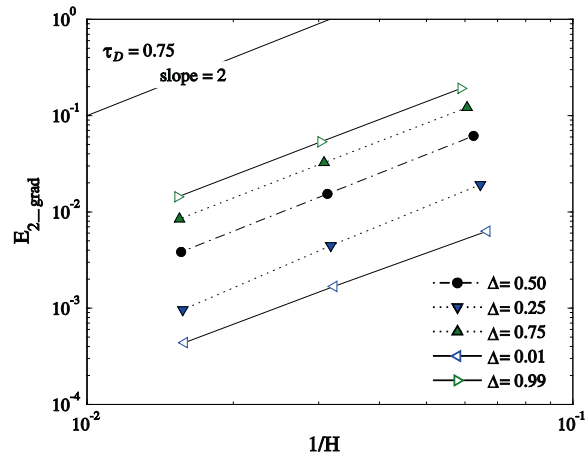


Fig. 15. Interior gradient error $E_{2,grad}$ versus the grid resolution, $1/H$, for the channel flow Neumann problem.

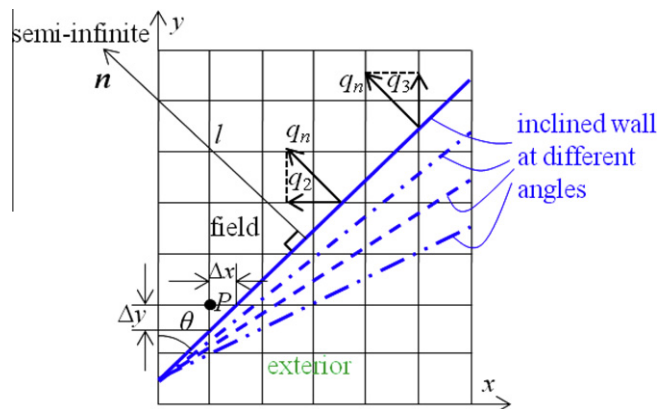


Fig. 16. Layout of the lattice around the inclined semi-infinite solid.

The initial temperature is zero in the field (solid), and the constant heat flux condition is imposed as

$$q_w = -D_t \frac{\partial T}{\partial n} \Big|_{l=0} = \frac{\sqrt{D_t \pi}}{2}. \quad (62)$$

Hence the transient temperature along the solid is solved to be

$$T_{\text{ex}}(t, l) = t^{1/2} \left[e^{-l^2/(4D_t t)} - \frac{l\sqrt{\pi}}{2\sqrt{D_t t}} \operatorname{erfc} \left(\frac{l}{2\sqrt{D_t t}} \right) \right], \quad (63)$$

where l is the distance in the normal direction and “erfc” is the complementary error function. The temperature on the wall is $f(t) = T(t, l=0) = t^{1/2}$.

With no temperature variation along the tangential direction, the boundary heat fluxes in the discrete velocity directions can be directly calculated using Eq. (49),

$$q_2 = q_w \cos \theta = \frac{\sqrt{D_t \pi}}{2} \cos \theta \quad (64a)$$

and

$$q_3 = q_w \sin \theta = \frac{\sqrt{D_t \pi}}{2} \sin \theta. \quad (64b)$$

Fig. 17(a) shows the variations of the temperature with time at point P in Fig. 16 for inclination angles $\theta = \tan^{-1}(1.0)$, $\tan^{-1}(1.2)$, $\tan^{-1}(1.5)$, and $\tan^{-1}(2.0)$. For each inclination angle, the temperature profile along the vertical line going through point P at $t = 500\delta t$ is compared with the exact solution in Fig. 17(b), where the distance in the y coordinate has been projected into that in the normal direction of the physical boundary.

Very good agreement between the simulation and the analytical solution is noticed in Fig. 17(a and b). These results verify that Eq. (49) is correct when the tangential temperature gradient on the boundary is zero. This test also serves as a good example to show that the present Neumann boundary condition treatment can be readily implemented for a boundary that is intersected by the rectangular lattice with arbitrary link fractions, i.e., irregularly spaced Δ values, which is the subject of the next example.

3.3. Two-D steady-state heat conduction inside a circle

Fig. 18(a) shows the schematic layout of the lattice covering the entire surface and the interior of a circle of radius r_0 . The specific Δ_x or Δ_y value of each boundary cut must be calculated first. It should be noted that the Δ values are irregularly distributed along the azimuthal direction; as an example, the distribution of Δ values, including both Δ_x and Δ_y , as a function of the azimuthal angle φ of the boundary nodes at $r_0 = 30.5$ is depicted in Fig. 18(b).

3.3.1. Dirichlet boundary condition

First we consider the steady-state Dirichlet boundary condition

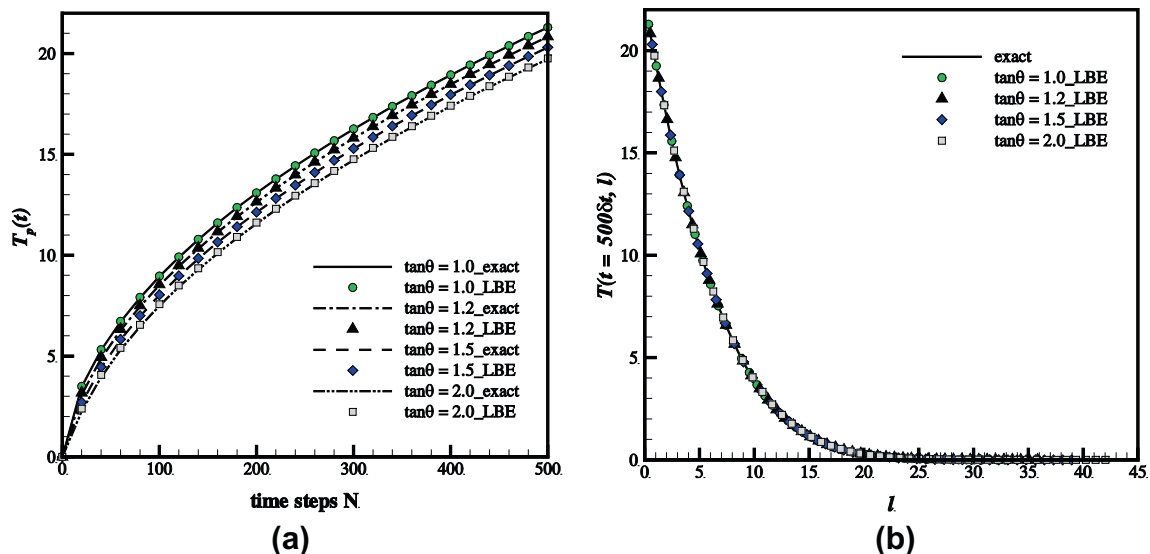


Fig. 17. (a) Temperature variations with time at point P ; (b) temperature profiles in the y -direction going through P at $t = 500\delta t$.

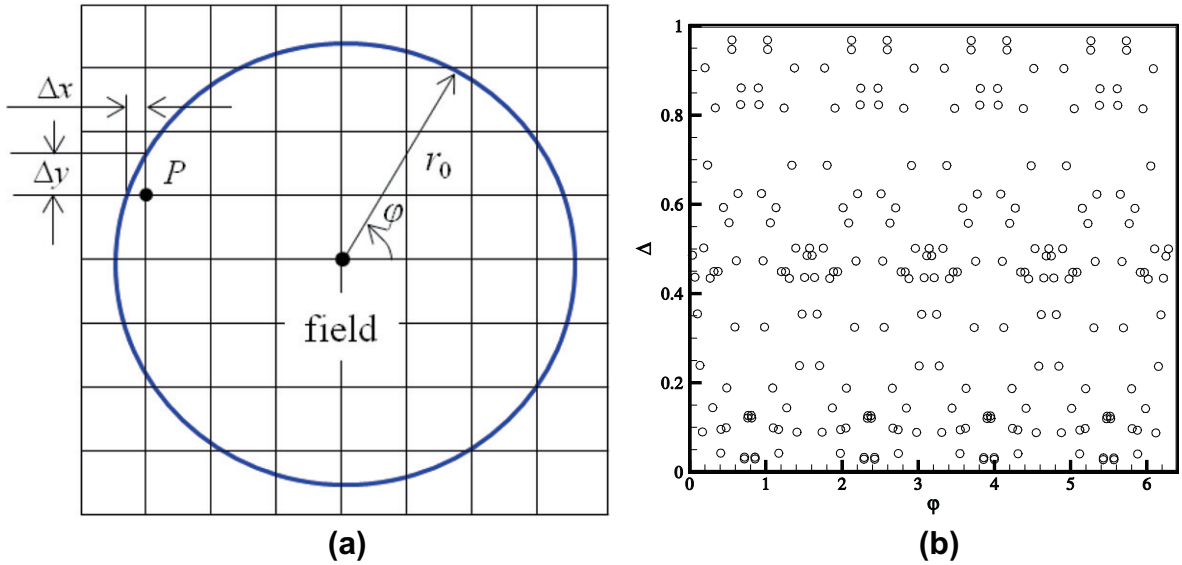


Fig. 18. (a) Schematic layout of the lattice around a circle; (b) distribution of Δ values along the azimuthal angle φ at $r_0 = 30.5$.

$$T(r = r_0) = f(\varphi) = \cos(n\varphi); \quad n = \text{integer}. \quad (65)$$

The exact solution for the temperature field is

$$T_{\text{ex}}(r, \varphi) = (r/r_0)^n \cos(n\varphi). \quad (66)$$

Fig. 19 shows the relative L_2 error norm of E_2 defined in Eq. (56) for all the interior nodes versus the grid resolution, $1/r_0$, on the log–log scale with $n = 4$ used in Eq. (65). The present Schemes 1, 2 and 3 are based on Eq. (35a–c), respectively, and the “third-order” and second-order schemes proposed by Ginzburg [48] are used for comparison (those schemes are rewritten in Eqs. (36), (37)). Second-order accuracy for the temperature field is observed in Fig. 19 for each scheme examined. It is seen that Scheme 2 gives better accuracy comparing with the other two given by Eq. (35).

The results for the error norm $E_{2,\text{qw}}$ of the boundary heat flux Φ_{nz} according to Eq. (41a) are shown in Fig. 20, where on average only first-order convergence is observed. It should be noted that in Fig. 20 the results of Ginzburg’s third- and second-order schemes are obtained by combining Eqs. (36) and (37) with our second-order Neumann scheme in Eq. (40), respectively, and following the same procedure as that for Eq. (41a). Fig. 21 gives the results of the error norm of the interior temperature gradient $E_{2,\text{grad}}$ defined in Eq. (58). Realizing Scheme 2 has better accuracy than the other two given by Eq. (35) in the previous examples only the results using Scheme 2 are presented in Figs. 20 and 21. The “finite-difference” case

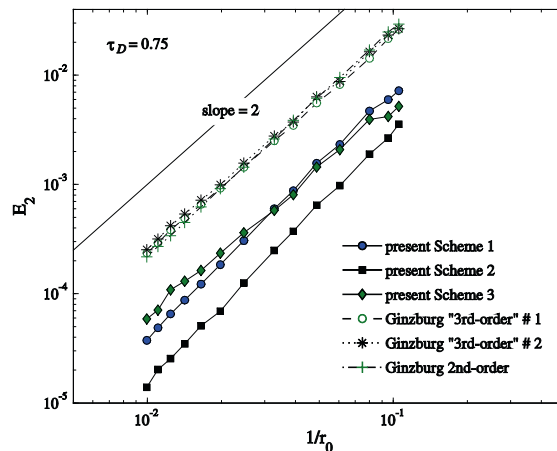


Fig. 19. Relative L_2 error norm E_2 versus the grid resolution, $1/r_0$, for the steady-state heat conduction inside a circle with a Dirichlet condition (results based on Ginzburg’s schemes [48] are included).

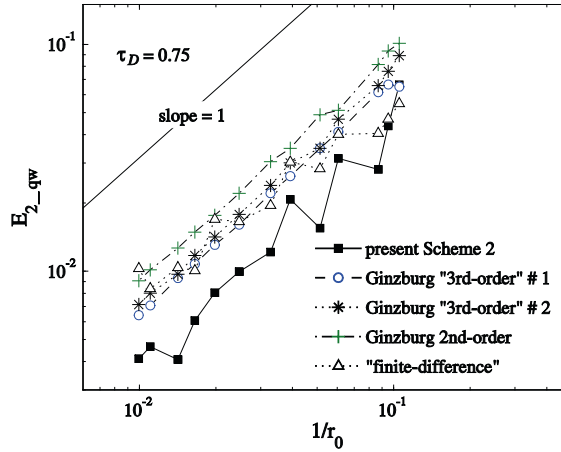


Fig. 20. Wall heat flux error $E_{2,qw}$ versus the grid resolution, $1/r_0$, for the steady-state heat conduction inside a circle with a Dirichlet condition (results based on Ginzburg's schemes [48] are included).

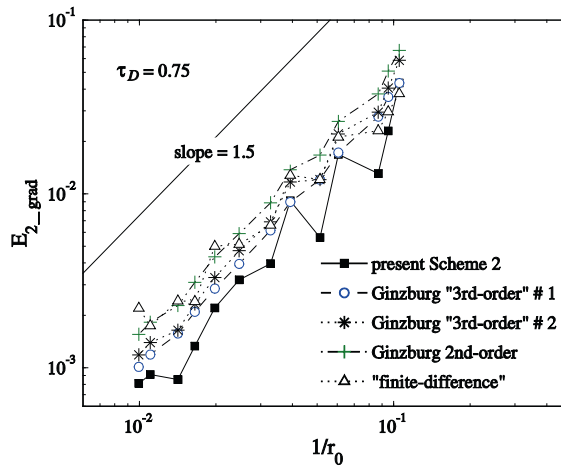


Fig. 21. Interior gradient error $E_{2,grad}$ versus the grid resolution, $1/r_0$, for the steady-state heat conduction inside a circle with a Dirichlet condition (results based on Ginzburg's schemes [48] are included).

represents a quadratic finite-difference scheme based on the numerical values of the temperature field at the lattice nodes and the given wall temperature.

One valuable insight to be gained from Figs. 20 and 21 is that for boundary heat flux and interior temperature gradient computed from Eqs. (41a) and (42), respectively, these various Dirichlet schemes all yield linear convergence for the boundary heat flux, and the error norm of the interior temperature gradient $E_{2,grad}$ is proportional to $(1/r_0)^{1.5}$, which are lower than the second-order convergence as shown in Figs. 10 and 11, respectively for straight boundaries. The degradation of the convergence orders for the derivative related physical quantities is due to the curved boundary on which the intersected link fractions (Δ values) are irregularly distributed (see Fig. 18(b)). On the whole, the present Dirichlet condition treatment on curved boundaries is able to produce second-order accurate temperature field and first-order boundary heat flux, and the interior temperature gradient converges to its exact solution with a “superlinear” order close to 1.5. This observation also offers an insight into the accuracy of the Neumann boundary condition treatment on curved boundaries, which will be discussed next.

3.3.2. Neumann boundary condition

Now consider the following Neumann (flux) condition applied on the boundary

$$q_w = -D_t \left. \frac{\partial T}{\partial r} \right|_{r_0} = -n D_t \cos(n\varphi) / r_0. \quad (67)$$

The solution of the temperature field is the same as that in Eq. (66) and the case of $n = 4$ is used.

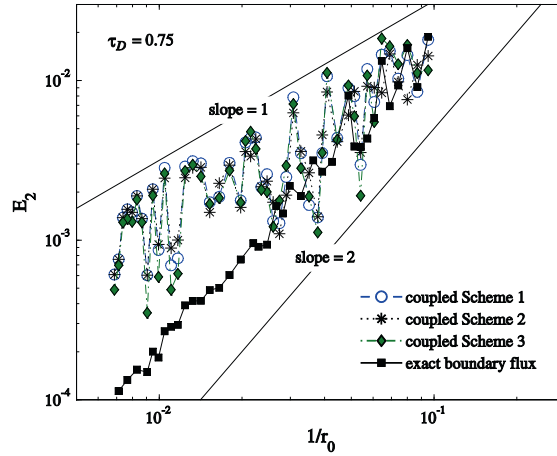


Fig. 22. Relative L_2 error norm E_2 versus the grid resolution, $1/r_0$, for the steady-state heat conduction inside a circle with a Neumann condition.

To account for the contribution to $\Phi_{n\bar{x}}$ from the tangential temperature gradient, Eq. (48) is used. After $\Phi_{n\bar{x}}$ is obtained numerically, Eq. (40) is implemented to complete the streaming step for Neumann problems. The error norms E_2 , E_{2_tw} and E_{2_grad} of the interior temperature field, the boundary temperature, and the interior temperature gradient as defined in Eqs. (56), (61), and (58) versus the grid resolution, $1/r_0$, are shown in Figs. 22–24, respectively, where the “coupled Schemes 1, 2 and 3” represent the different coefficient sets c_{di} and c'_{di} in Eq. (46) determined by the three schemes given by Eq. (35a–c), respectively.

From the analytical solution for the temperature field, Eq. (66), the exact boundary heat flux $\Phi_{n\bar{x}}$ in the discrete velocity direction $\mathbf{e}_{\bar{x}}$ can be obtained with projections of the normal and tangential heat fluxes into the x - or y -coordinate. For comparison, the results of the three types of error norms E_2 , E_{2_tw} and E_{2_grad} based on this exact boundary flux are also presented in Figs. 22–24, respectively.

First of all, we notice that with the “exact boundary flux” $\Phi_{n\bar{x}}$ applied, the use of Eq. (40) for evaluation of the distribution function leads to second-order accuracy for all the three quantities, i.e., the interior and boundary temperatures and the interior temperature gradients. However, only first-order accuracy is obtained for both the interior and boundary temperatures for all the three “coupled” schemes. The average decrease of the error norms in Figs. 22 and 23 is proportional to the grid resolution, $1/r_0$, on the log–log scale and considerable oscillations are observed even at high grid resolution. The interior temperature gradient in Fig. 24 still converges with a “superlinear” order around 1.5, which is similar to the result of E_{2_grad} in Fig. 21 with a Dirichlet condition. Compared with the second-order convergence in the case of exact boundary flux, the degradation of order-of-accuracy in the coupled schemes is due to the heat flux approximation of $\Phi_{n\bar{x}}$ for irregularly distributed Δ values along the curved boundary. Recalling in Fig. 20 only a linear convergence of the boundary heat flux is obtained for the Dirichlet problem, and the heat flux formula Eq. (41a), (46a) is used in deriving the ultimate heat flux approximation in Eq. (48), it is believed that some second-order errors terms $O(\delta_x^2)$, which are related to the local Δ values, are introduced in the heat flux approximation. Those second-order errors cannot always be cancelled out due to the “random” distribution of

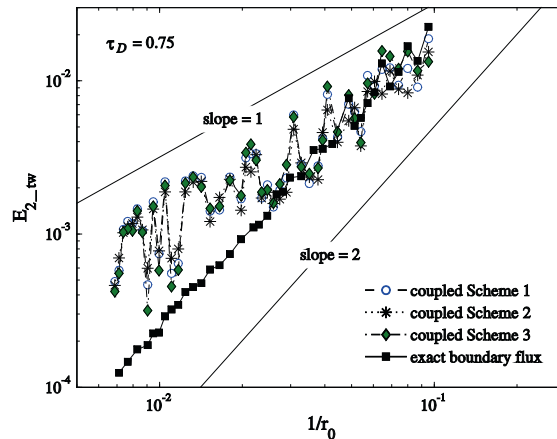


Fig. 23. Wall temperature error E_{2_tw} versus the grid resolution, $1/r_0$, for the steady-state heat conduction inside a circle with a Neumann condition.

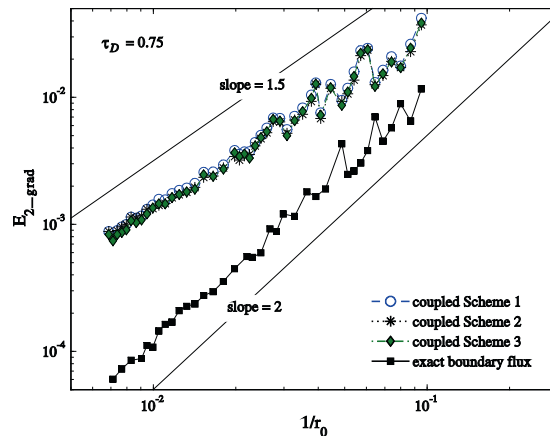


Fig. 24. Interior gradient error $E_{2,grad}$ versus the grid resolution, $1/r_0$, for the steady-state heat conduction inside a circle with a Neumann condition.

Δ values (Fig. 18 (b)), resulting in averaged linear convergence and persistent oscillations even at high resolution as shown in Figs. 22 and 23. Based on the “multi-reflection” approach, Ginzburg [48] also found that only linear convergence can be achieved for the Neumann condition on curved boundaries when tangential temperature gradient exists.

The orders of accuracy for the Dirichlet and Neumann conditions on straight and curved boundaries for the temperature and its derivative related quantities are summarized in Table 1 for 2-D steady thermal problems.

3.4. Two-D transient heat conduction inside a circle

In this case, a transient boundary condition is applied on the curved boundary to examine the temporal accuracy of the present boundary condition treatment for time-dependent problems. The geometry is the same as that in Fig. 18(a).

The temperature is assumed uniform in the azimuthal direction and the transient Dirichlet boundary condition reads

$$T(t, r = r_0) = f(t) = \sin(\omega t), \quad (68)$$

where ω is the frequency of the imposed thermal boundary condition. This transient problem is characterized by the Stokes number St defined as

$$St^2 = \frac{r_0^2 \omega}{2\pi D_t}. \quad (69)$$

The analytical solution for this 2-D problem can be found as [49]

$$\begin{aligned} T_{ex}(r, t) &= f(t) - \frac{2}{r_0} \sum_{n=1}^{\infty} \frac{J_0(\beta_n r)}{\beta_n J_1(\beta_n r_0)} \left[f(0) e^{-D_t \beta_n^2 t} + \int_0^t e^{-D_t \beta_n^2 (t-\tau)} df(\tau) \right] \\ &= \sin(\omega t) - \frac{2}{r_0} \sum_{n=1}^{\infty} \frac{\omega J_0(\beta_n r)}{\beta_n J_1(\beta_n r_0)} \left[\frac{D_t \beta_n^2 \cos(\omega t) + \omega \sin(\omega t) - D_t \beta_n^2 e^{-D_t \beta_n^2 t}}{(D_t \beta_n^2)^2 + \omega^2} \right], \end{aligned} \quad (70)$$

where $J_i(\beta r)$ ($i = 0, 1$) is the solution to the Bessel's function of the first kind of order i , and β_n is determined by $J_0(\beta_n r_0) = 0$.

For error assessment purposes we define the following error norms

Table 1
Accuracy of the TLBE simulations with Dirichlet/Neumann conditions.

	Dirichlet condition		Neumann condition		
	Parallel straight boundary [†]	Curved boundary	Parallel straight boundary [†]	Curved boundary [§]	Curved boundary [¶]
Interior temperature field	2nd-order	2nd-order	2nd-order	2nd-order	1st-order
Boundary heat flux	2nd-order	1st-order	—	—	—
Boundary temperature	—	—	2nd-order	2nd-order	1st-order
Interior temperature gradient	2nd-order	“superlinear” order ~1.5	2nd-order	2nd-order	“superlinear” order ~1.5

[†]The boundary normal is parallel with a lattice vector. [§]Exact boundary heat flux. [¶]Coupled schemes.

$$E(t^*) = \left[\frac{1}{N_x N_y} \sum_{x,y} (T_{\text{LBE}} - T_{\text{ex}})^2 \right]^{1/2}, \quad (71)$$

$$E_2 = \left[\frac{\omega}{2\pi} \int_0^{2\pi/\omega} \frac{1}{N_x N_y} \sum_{x,y} (T_{\text{LBE}} - T_{\text{ex}})^2 dt \right]^{1/2} = \left[\int_0^1 \frac{1}{N_x N_y} \sum_{x,y} (T_{\text{LBE}} - T_{\text{ex}})^2 dt^* \right]^{1/2}, \quad (72)$$

where $t^* = \omega t / 2\pi$ and the product of $(N_x N_y)$ accounts for the total number of lattice nodes inside the circle.

To eliminate the effect of initial transient, the computation is performed for a sufficiently long period of time ($t^* > 1$). The Stokes number St is set to be unity in all simulations. The results of $E(t^*)$ defined in Eq. (71) at $\tau_D = 0.51$ and 0.75 are presented in Fig. 25 using the three schemes in Eq. (35a–c). The grid resolution here is relatively poor as $r_0 = 5.8$. The average error E_2 defined in Eq. (72) is evaluated over one single period after the dynamic steady state is reached. The results of E_2 with different resolution are shown in Fig. 26 as a function of $1/r_0$ for $\tau_D = 0.75$. The decrease of the error norm E_2 is proportional to $(1/r_0)^2$ on the log–log scale for each of the three schemes. It can also be observed in Fig. 26 that Schemes 1 and 2 have very close errors and they are more accurate than Scheme 3 in general. When $\tau_D = 0.51$ is used, the three schemes give almost identical results for E_2 and the magnitude is very close to that in Fig. 26 for $\tau_D = 0.75$ thus they are not shown here for brevity. Overall, Scheme 2 appears to give smaller error than the other two schemes given by Eq. (35).

It is thus confirmed that the present Dirichlet boundary condition treatment can be applied in transient problems with curved-geometry, and second-order accuracy is preserved for the temperature field.

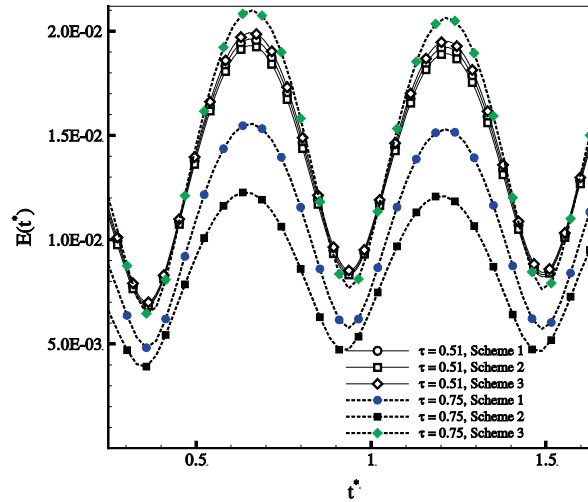


Fig. 25. Comparison of the error norm $E(t^*)$ defined in Eq. (71) for the transient heat conduction inside a circle using the three schemes with $r_0 = 5.8$.

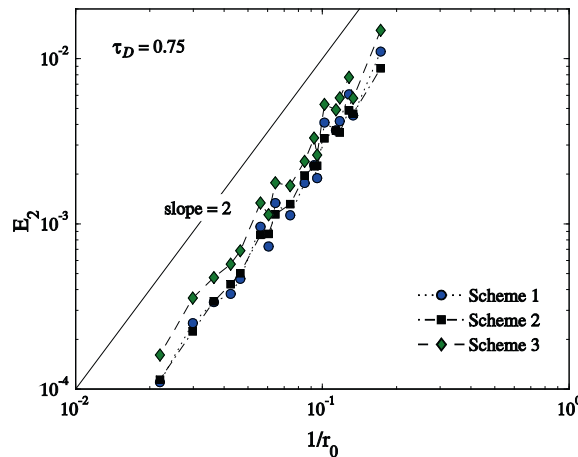


Fig. 26. Relative L_2 error norm E_2 versus the grid resolution, $1/r_0$, for the transient heat conduction inside a circle.

3.5. Three-D steady-state circular pipe flow

To test the TLBE model and the boundary condition treatments in 3-D cases, the convection and diffusion inside a circular pipe are considered in this example. The temperature is assumed to be uniform in the azimuthal direction in order to obtain analytical solutions.

3.5.1. Dirichlet boundary condition

A plug flow with constant velocity U in the axial z -direction is assumed. The periodic boundary condition in the z -direction is applied as

$$T(r, z = L) = T(r, z = 0), \quad (73)$$

and the Dirichlet boundary condition is imposed as

$$T(r = r_0, z) = \cos(kz), \text{ with } k = 2\pi/L. \quad (74)$$

Following the same procedure for the 2-D channel flow with Dirichlet boundary conditions in Section 3.1.1, the analytical solution for the temperature field is solved to be

$$T_{\text{ex}}(r, z) = \text{Real} \left[e^{ikz} \frac{I_0(\lambda r)}{I_0(\lambda r_0)} \right], \quad (75)$$

where $I_0(\lambda r)$ is the modified Bessel's function of the first kind of order 0 and $\lambda = k \sqrt{1 + \frac{iU}{D_t k}}$.

The relative error norms E_2 and $E_{2-\text{qw}}$ are extended from 2-D to 3-D cases

$$E_2 = \left[\sum_{r, \theta, z} (T_{\text{LBE}} - T_{\text{ex}})^2 / \sum_{r, \theta, z} T_{\text{ex}}^2 \right]^{1/2}, \quad (76)$$

$$E_{2-\text{qw}} = \left[\sum_{r=r_0, \theta, z} \left(\left(\frac{\partial T}{\partial r} \right)_{\text{LBE}} - \left(\frac{\partial T}{\partial r} \right)_{\text{ex}} \right)^2 / \sum_{r=r_0, \theta, z} \left(\frac{\partial T}{\partial r} \right)_{\text{ex}}^2 \right]^{1/2}. \quad (77)$$

Figs. 27 and 28 show the results of E_2 and $E_{2-\text{qw}}$ for this 3-D case, respectively, versus the grid resolution, $1/r_0$, with the three schemes in Eq. (35a–c) implemented. The characteristic Péclet number is $Pe = (2r_0)U/D_t = 20$ and the relaxation coefficient is $\tau_D = 0.75$.

Similar to the 2-D example in Section 3.1.1, this 3-D example with a Dirichlet condition on the curved boundary gives quadratic convergence of the temperature field and linear convergence of the boundary heat flux. The second-order accuracy of the present Dirichlet boundary condition treatment for temperature fields is therefore verified for 3-D thermal flows with curved geometry.

3.5.2. Neumann boundary condition

For the Neumann problem, the heat flux is given as

$$q_w = -D_t \frac{\partial T}{\partial r} \Big|_{r=r_0} = -D_t \cos(kz)/r_0, \quad (78)$$

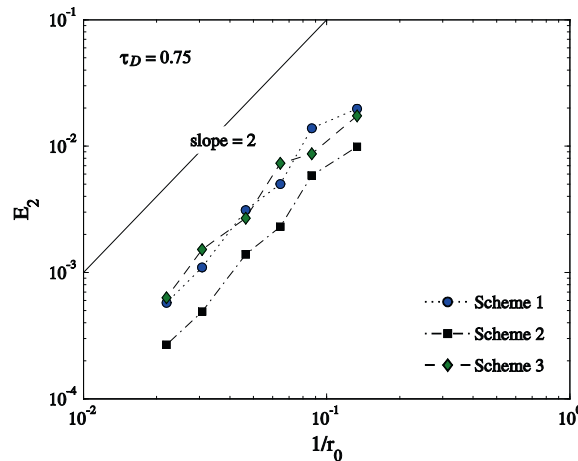


Fig. 27. Relative L-2 error norm E_2 versus the grid resolution, $1/r_0$, for the pipe flow with a Dirichlet condition.

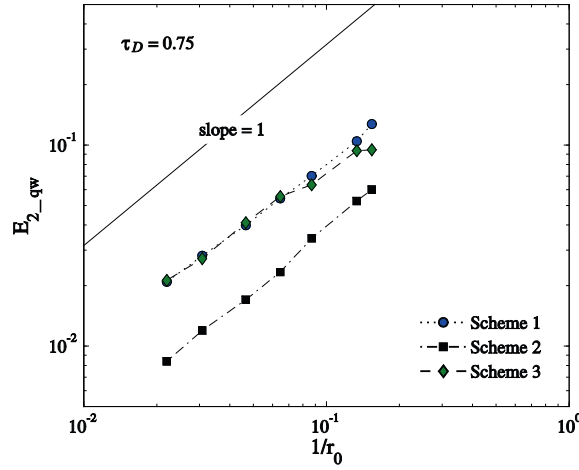


Fig. 28. Wall heat flux error $E_{2,qw}$ versus the grid resolution, $1/r_0$, for the pipe flow with a Dirichlet condition.

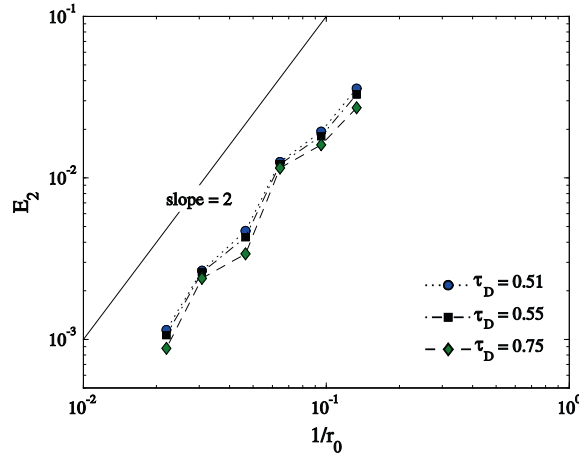


Fig. 29. Relative L_2 error norm E_2 versus the grid resolution, $1/r_0$, for the pipe flow with a Neumann condition.

and other conditions are the same as in Section 3.5.1. The analytical solution for the temperature field is

$$T_{ex}(r, z) = \text{Real} \left[e^{ikz} \frac{I_0(\lambda r)}{\lambda r_0 I_1(\lambda r_0)} \right], \quad (79)$$

where $I_1(\lambda r)$ is the modified Bessel's function of the first kind of order 1. There is no azimuthal temperature gradient along the curved boundary; the non-zero tangential gradient is on the z -direction, which is aligned with $\mathbf{e}_5$ and $\mathbf{e}_6$ of the lattice velocity vectors. Thus the exact flux conversion formula in Eq. (49) can be directly applied.

The error norm E_2 defined in Eq. (76) is again examined for the Neumann problem and the results at $\tau_D = 0.51, 0.55$ and 0.75 are shown in Fig. 29.

For each of the τ_D values tested, the present Neumann condition treatment leads to a stable computation. The decrease of the relative error E_2 is proportional to $(1/r_0)^2$ on the log-log scale. The temperature field obtained is thus second-order accurate. Comparing with the linear convergence of the temperature field in Section 3.3.2, it is confirmed that with zero temperature gradient along the curved boundary, second-order accuracy of temperature fields can be achieved with the present Neumann boundary condition treatment for 2-D and 3-D problems.

4. Conclusions

Thermal boundary condition treatments based on the “bounce-back” scheme and spatial interpolation are presented for both the Dirichlet and Neumann conditions. When second-order accuracy is pursued, there is an adjustable parameter for the Dirichlet problem, while the Neumann condition treatment is unique. Three particular schemes for the Dirichlet condition treatment are presented and their stability and accuracy are examined. All three Dirichlet schemes are stable and Scheme 2 gives smaller error in most cases investigated.

When extended to curved boundaries, the present Dirichlet condition treatment can be applied directly and it leads to second-order accuracy of the temperature field and first-order accuracy of the boundary heat flux. In contrast the proposed Neumann condition treatment requires the determination of heat fluxes in the discrete velocity directions of the TLBE model. A general formula to convert the normal heat flux into the fluxes in the discrete velocity directions is derived. Different from the Dirichlet problem, only first-order accuracy is obtained for the temperature field with Neumann conditions due to the combined effect of the curved boundary and the heat flux conversion. For the special case with no tangential variation of temperature along the curved boundary, the heat flux conversion becomes exact and second-order accuracy of the temperature field can be obtained.

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Appendix. Asymptotic analysis of second-order accurate boundary schemes

The asymptotic analysis and diffusive scaling in [28] are applied here with the following dimensionless parameter introduced

$$\epsilon = \delta x / L, \quad (\text{A1})$$

where L is the reference length. For details of the asymptotic analysis of the thermal LBE and the diffusive scaling please refer to the work in [28]. Here we focus on the asymptotic analysis for the present boundary condition treatments; some results given in [28] are used directly.

The distribution function g_α and the macroscopic variable ϕ can be expanded as (the arguments of $(\mathbf{x}, t)$ are dropped for simplicity)

$$g_\alpha = g_\alpha^{(0)} + \epsilon g_\alpha^{(1)} + \epsilon^2 g_\alpha^{(2)} + O(\epsilon^3), \quad (\text{A2})$$

$$\phi = \phi^{(0)} + \epsilon \phi^{(1)} + \epsilon^2 \phi^{(2)} + O(\epsilon^3), \quad (\text{A3})$$

and the spatial vectors $\mathbf{x}_f$ and $\mathbf{x}_{ff}$ in the field are related to the boundary vector $\mathbf{x}_w$ by (see Fig. 2)

$$\mathbf{x}_f = \mathbf{x}_w - \Delta \mathbf{e}_x \epsilon = \mathbf{x}_w + \Delta \mathbf{e}_{\bar{x}} \epsilon, \quad (\text{A4a})$$

$$\mathbf{x}_{ff} = \mathbf{x}_w - (1 + \Delta) \mathbf{e}_x \epsilon. \quad (\text{A4b})$$

Please note that the spatial vector $\mathbf{x}$ and the time t throughout this Appendix are dimensionless variables normalized by the reference length and diffusion time scales, respectively. Based on the diffusive scaling in [28] the present boundary condition treatments in Eqs. (32) and (38) can be rewritten in terms of $\mathbf{x}_w$ as follows

Dirichlet boundary condition:

$$g_{\bar{x}}(\mathbf{x}_w + \Delta \mathbf{e}_{\bar{x}} \epsilon, t + \epsilon^2) = c_{d1} \hat{g}_x(\mathbf{x}_w - \Delta \mathbf{e}_x \epsilon, t) + c_{d2} \hat{g}_x(\mathbf{x}_w - (1 + \Delta) \mathbf{e}_x \epsilon, t) + c_{d3} \hat{g}_{\bar{x}}(\mathbf{x}_w + \Delta \mathbf{e}_{\bar{x}} \epsilon, t) + c_{d4} \epsilon_D \Phi_d, \quad (\text{A5})$$

and

Neumann boundary condition:

$$g_{\bar{x}}(\mathbf{x}_w + \Delta \mathbf{e}_{\bar{x}} \epsilon, t + \epsilon^2) = c_{n1} \hat{g}_x(\mathbf{x}_w - \Delta \mathbf{e}_x \epsilon, t) + c_{n2} \hat{g}_x(\mathbf{x}_w - (1 + \Delta) \mathbf{e}_x \epsilon, t) + c_{n3} \hat{g}_{\bar{x}}(\mathbf{x}_w + \Delta \mathbf{e}_{\bar{x}} \epsilon, t) + c_{n4} \epsilon \Phi_{n\bar{x}} / U, \quad (\text{A6})$$

where U is a reference speed ($U = \epsilon \delta x / \delta t$) for diffusive scaling as defined in [28]. The source term does not affect boundary condition treatment; thus it is not considered here for simplicity. The post-collision distribution functions in Eqs. (A5), (A6) are manipulated with the collision operator, and after inserting the expansion (A2) into Eqs. (A5), (A6) each distribution can be expressed with respect to the boundary distribution function, $g_\alpha^{(m)}(\mathbf{x}_w, t)$, or $g_\alpha^{(m)}(\mathbf{x}_w, t)$ ($m = 0, 1$, and 2) using Taylor-series expansions as below

$$g_\alpha^{(m)}(\mathbf{x}_w + \Delta \mathbf{e}_{\bar{x}} \epsilon, t + \epsilon^2) = g_\alpha^{(m)} + \left(\Delta \epsilon e_{xj} \frac{\partial}{\partial x_j} + \epsilon^2 \frac{\partial}{\partial t} + \frac{1}{2} \Delta^2 \epsilon^2 e_{xi} e_{xj} \frac{\partial^2}{\partial x_i \partial x_j} + O(\epsilon^3) \right) g_\alpha^{(m)}, \quad (\text{A7a})$$

$$\begin{aligned} \hat{g}_x^{(m)}(\mathbf{x}_w - \Delta \mathbf{e}_x \epsilon, t) &= g_x^{(m)}(\mathbf{x}_w - \Delta \mathbf{e}_x \epsilon, t) + \mathbf{L} g_x^{(m)}(\mathbf{x}_w - \Delta \mathbf{e}_x \epsilon, t) \\ &= (g_x^{(m)} + \mathbf{L} g_x^{(m)}) + \left(-\Delta \epsilon e_{xj} \frac{\partial}{\partial x_j} + \frac{1}{2} \Delta^2 \epsilon^2 e_{xi} e_{xj} \frac{\partial^2}{\partial x_i \partial x_j} + O(\epsilon^3) \right) (g_x^{(m)} + \mathbf{L} g_x^{(m)}), \end{aligned} \quad (\text{A7b})$$

$$\begin{aligned}\hat{g}_\alpha^{(m)}(\mathbf{x}_w - (1 + \Delta)\mathbf{e}_\alpha\epsilon, t) &= g_\alpha^{(m)}(\mathbf{x}_w - (1 + \Delta)\mathbf{e}_\alpha\epsilon, t) + \mathbf{L}g_\alpha^{(m)}(\mathbf{x}_w - (1 + \Delta)\mathbf{e}_\alpha\epsilon, t) \\ &= (g_\alpha^{(m)} + \mathbf{L}g_\alpha^{(m)}) + \left(-(1 + \Delta)\epsilon e_{\alpha j} \frac{\partial}{\partial x_j} + \frac{1}{2}(1 + \Delta)^2 \epsilon^2 e_{\alpha i} e_{\alpha j} \frac{\partial^2}{\partial x_i \partial x_j} + O(\epsilon^3) \right) (g_\alpha^{(m)} + \mathbf{L}g_\alpha^{(m)}),\end{aligned}\quad (\text{A7c})$$

$$\begin{aligned}\hat{g}_\alpha^{(m)}(\mathbf{x}_w + \Delta\mathbf{e}_\alpha\epsilon, t) &= g_\alpha^{(m)}(\mathbf{x}_w + \Delta\mathbf{e}_\alpha\epsilon, t) + \mathbf{L}g_\alpha^{(m)}(\mathbf{x}_w + \Delta\mathbf{e}_\alpha\epsilon, t) \\ &= (g_\alpha^{(m)} + \mathbf{L}g_\alpha^{(m)}) + \left(\Delta\epsilon e_{\alpha j} \frac{\partial}{\partial x_j} + \frac{1}{2}\Delta^2 \epsilon^2 e_{\alpha i} e_{\alpha j} \frac{\partial^2}{\partial x_i \partial x_j} + O(\epsilon^3) \right) (g_\alpha^{(m)} + \mathbf{L}g_\alpha^{(m)}),\end{aligned}\quad (\text{A7d})$$

where the arguments $(\mathbf{x}_w, t)$ on the right of the equations are skipped for brevity and the collision operator is (see Eq. (9))

$$\mathbf{L}g_\alpha^{(m)} = \mathbf{M}^{-1}\mathbf{SM}(\mathbf{Q}^{(0)} + \epsilon\mathbf{Q}^{(1)})g_\alpha^{(m)}, \quad (\text{A8})$$

with

$$\mathbf{Q}^{(0)} = (\omega_0, \omega_1, \dots, \omega_6)^T (1, 1, \dots, 1) - \mathbf{I}_{7 \times 7}, \quad (\text{A9a})$$

and

$$\mathbf{Q}^{(1)} = \frac{1}{\varepsilon_D U} (0, u\omega_1, -u\omega_2, v\omega_3, -v\omega_4, w\omega_5, -w\omega_6)^T (1, 1, 1, 1, 1, 1, 1), \quad (\text{A9b})$$

Substitution of Eqs. (A7a–A7d) into the boundary schemes given by Eqs. (A5), (A6) and by rearranging the relevant terms according to the power of ϵ , one can obtain the required conditions to satisfy the Dirichlet and Neumann boundary conditions to second-order accuracy.

Dirichlet boundary condition:

The equation of zeroth-order in ϵ is

$$(1 - c_{d3})g_\alpha^{(0)} - (c_{d1} + c_{d2})g_\alpha^{(0)} = c_{d4}\varepsilon_D\Phi_d. \quad (\text{A10})$$

In the power of ϵ^1 , the equation is

$$(1 - c_{d3})g_\alpha^{(1)} - (c_{d1} + c_{d2})g_\alpha^{(1)} + \left[(\Delta(1 - c_{d3}) - c_{d3})e_{\alpha j} \frac{\partial g_\alpha^{(0)}}{\partial x_j} + ((\Delta - 1)(c_{d1} + c_{d2}) + c_{d2})e_{\alpha j} \frac{\partial g_\alpha^{(0)}}{\partial x_j} \right] = 0, \quad (\text{A11})$$

According to Yoshida and Nagaoka [28], the following relationships are noticed based on Eqs. (50), (51), and (57) in [28], respectively

$$g_\alpha^{(0)} - g_\alpha^{(0)} = 0, \quad (\text{A12})$$

$$g_\alpha^{(0)} + g_\alpha^{(0)} = \varepsilon_D\phi^{(0)}, \quad (\text{A13})$$

$$g_\alpha^{(1)} + g_\alpha^{(1)} = \varepsilon_D\phi^{(1)}. \quad (\text{A14})$$

Eqs. (A12–A14) imply that the following relationships must hold for the coefficients in Eqs. (A10) and (A11)

$$\begin{cases} 1 - c_{d3} = -(c_{d1} + c_{d2}) \\ 1 - c_{d3} = c_{d4} \\ \Delta(1 - c_{d3}) - c_{d3} = (\Delta - 1)(c_{d1} + c_{d2}) + c_{d2} \\ c_{d4} \neq 0, \end{cases} \quad (\text{A15})$$

in order to satisfy

$$\phi^{(0)} = \Phi_d, \quad (\text{A16})$$

and

$$\phi^{(1)} = 0. \quad (\text{A17})$$

Noting that the Dirichlet thermal boundary condition is given by

$$\phi = \Phi_d, \quad (2)$$

and recalling the expansion $\phi = \phi^{(0)} + \epsilon\phi^{(1)} + \epsilon^2\phi^{(2)} + O(\epsilon^3)$ in (A3), the following is thus obtained

$$\phi^{(0)} = \Phi_d + O(\epsilon^2). \quad (\text{A18})$$

Eq. (A18) verifies that the Dirichlet boundary condition is satisfied up to second-order accuracy with the leading-order solution solved by $\phi^{(0)}$ in TLBE simulations. The coefficients in Eq. (A15) cannot be uniquely determined, and leaving c_{d1} adjustable the constraints can be rewritten as

$$\begin{cases} c_{d2} = -\frac{2\Delta c_{d1}+1}{2\Delta+1} \\ c_{d3} = \frac{c_{d1}+2\Delta}{2\Delta+1} \\ c_{d4} = \frac{-c_{d1}+1}{2\Delta+1} \\ c_{d1} \neq 1. \end{cases} \quad (\text{A19})$$

Neumann boundary condition:

The Neumann boundary condition is given by Eq. (3). For brevity, we assume the normal direction of the physical boundary is in the same direction as the discrete velocity direction $\mathbf{e}_\alpha$; thus $\Phi_{n\alpha} = \Phi_n$ can be directly applied.

Substitution of expansions (A7a–A7d) into the boundary scheme in (A6) gives the equation of zeroth-order in ϵ as

$$(1 - c_{n3})\mathbf{g}_\alpha^{(0)} - (c_{n1} + c_{n2})\mathbf{g}_\alpha^{(0)} = 0, \quad (\text{A20})$$

In the power of ϵ^1 , the equation is

$$(1 - c_{n3})\mathbf{g}_\alpha^{(1)} - (c_{n1} + c_{n2})\mathbf{g}_\alpha^{(1)} + \left[(\Delta(1 - c_{n3}) - c_{n3})\mathbf{e}_{\alpha j} \frac{\partial \mathbf{g}_\alpha^{(0)}}{\partial x_j} - ((1 - \Delta)(c_{n1} + c_{n2}) - c_{n2})\mathbf{e}_{\alpha j} \frac{\partial \mathbf{g}_\alpha^{(0)}}{\partial x_j} \right] = c_{n4} \frac{\Phi_n}{U}, \quad (\text{A21})$$

and proceeding further to $O(\epsilon)^2$ yields

$$\begin{aligned} & [(1 - c_{n3})\mathbf{g}_\alpha^{(2)} - (c_{n1} + c_{n2})\mathbf{g}_\alpha^{(2)}] + \left[(\Delta - c_{n3}(1 + \Delta))\mathbf{e}_{\alpha j} \frac{\partial \mathbf{g}_\alpha^{(1)}}{\partial x_j} - (c_{n1} - (c_{n1} + c_{n2})\Delta)\mathbf{e}_{\alpha j} \frac{\partial \mathbf{g}_\alpha^{(1)}}{\partial x_j} \right] \\ & + \left[\frac{1}{2}(\Delta^2 - c_{n3}(\Delta + 1)^2)\mathbf{e}_{\alpha i}\mathbf{e}_{\alpha j} \frac{\partial^2 \mathbf{g}_\alpha^{(0)}}{\partial x_i \partial x_j} - \frac{1}{2}(c_{n1}(\Delta - 1)^2 + c_{n2}\Delta^2)\mathbf{e}_{\alpha i}\mathbf{e}_{\alpha j} \frac{\partial^2 \mathbf{g}_\alpha^{(0)}}{\partial x_i \partial x_j} \right] \\ & = 0, \end{aligned} \quad (\text{A22})$$

With the expansion $\phi = \phi^{(0)} + \epsilon \phi^{(1)} + \epsilon^2 \phi^{(2)} + O(\epsilon^3)$ in (A3) one would expect that

$$-D_{ij} \frac{\partial \phi^{(0)}}{\partial x_j} + v_i \phi^{(0)} = \Phi_n, \quad (\text{A23})$$

so that the leading-order solution of $\phi^{(0)}$ satisfies the Neumann boundary condition. Recalling the relationship between D_{ij} and τ_{ij} in Eq. (14) the above (A23) can be rewritten as

$$\left(-\tau_{ij} + \frac{1}{2}\delta_{ij} \right) \epsilon_D \frac{\partial \phi^{(0)}}{\partial x_j} + \frac{v_i}{U} \phi^{(0)} = \frac{\Phi_n}{U}. \quad (\text{A24})$$

Besides Eqs. (A12–A14), the following relationships can be obtained from Eqs. (56) and (64) in [28], respectively

$$\mathbf{g}_\alpha^{(1)} - \mathbf{g}_\alpha^{(1)} = \frac{v_i}{U} \phi^{(0)} - \epsilon_D \tau_{ij} \frac{\partial \phi^{(0)}}{\partial x_j}, \quad (\text{A25})$$

$$\mathbf{g}_\alpha^{(2)} - \mathbf{g}_\alpha^{(2)} = \frac{v_i}{U} \phi^{(1)} - \epsilon_D \tau_{ij} \frac{\partial \phi^{(1)}}{\partial x_j}. \quad (\text{A26})$$

Comparing Eq. (A20) with Eq. (A12) and combining Eqs. (A13, A21, A24, and A25) we obtain the requirements

$$\begin{cases} 1 - c_{n3} = c_{n1} + c_{n2} \\ \Delta(1 - c_{n3}) - c_{n3} = (1 - \Delta)(c_{n1} + c_{n2}) - c_{n2} \\ 1 - c_{n3} = 2[\Delta(1 - c_{n3}) - c_{n3}] \\ c_{n4} = 1 - c_{n3}, \end{cases} \quad (\text{A27})$$

The unique solution to the coefficients in Eq. (A27) is

$$\begin{cases} c_{n1} = 1 \\ c_{n2} = -\frac{2\Delta-1}{2\Delta+1} \\ c_{n3} = \frac{2\Delta-1}{2\Delta+1} \\ c_{n4} = \frac{2}{2\Delta+1} \end{cases} \quad (\text{A28})$$

With the above coefficients and the relationships in Eqs. (A14) and (A26) the following can be derived from Eq. (A22)

$$\left(-\tau_{ij} + \frac{1}{2}\delta_{ij}\right)\varepsilon_D \frac{\partial \phi^{(1)}}{\partial x_j} + \frac{v_i}{U}\phi^{(1)} = 0. \quad (\text{A29})$$

As discussed in [28], since the condition for $\phi^{(1)}$ is linear and homogeneous, $\phi^{(1)}$ in (A29) has only a trivial solution of $\phi^{(1)} \equiv 0$. Thus we have verified that the Neumann boundary condition is satisfied to second-order accuracy.

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